The effect of child support on selection into marriage and fertility

by

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Abstract

Child support policies in the United States have expanded dramatically since the mid-1970s and now cover 1 in 5 children. This paper studies the consequences of child support for marriage and fertility decisions. I first introduce a model showing that child support enforces ex ante commitment from men to provide financial support in the event of a child, which (1) increases premarital sex among couples unlikely to marry, and (2) reduces the abortion rate, by lessening the cost of raising a child as a single mom. Using variation in the timing and geography of the rollout of U.S. child support laws relative to the timing of pregnancy, from 1977 to 1992, I find that marriages following an unplanned pregnancy are less likely to occur under strengthened child support laws, accounting for about a 7-8 percentage point reduction relative to a base of 38 percent. I find that the child support rollout reduced the abortion rate by 1-2 per 1000 women aged 15-44, off a base of 28, representing about 50 percent of the total decline in the abortion rate over this period.

Keyword*:* child support, family, marriage, abortion

JEL Classification: J12, J13, K36

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1 Introduction

In the last forty years the U.S. has dramatically strengthened the enforcement and collection of child support from non-resident parents. Beginning in 1975 with Part D of the Social Security Act, and continuing with major legislation in the 1980s and 1990s, these statutes were enacted to protect children and ensure that non-resident parents, the majority of whom are male, share in the burden of child rearing.[1]

Child support laws also have the potential to affect individual decisions regarding marriage and fertility. Child support laws ensure that mothers can receive financial support for child rearing without having to marry the father; they also allow fathers to have a legal relationship with their child without marriage. In addition, child support laws increase the cost of fathering a child, particularly an unwanted child, and males may respond by reducing their fertility. Understanding the impact of child support laws on family structure is of critical importance, not least because dramatic changes in marital patterns have coincided with a period of rapid child support expansion (Stevenson and Wolfers, 2007). To the extent that child support laws affect marital formation, they also affect the family structure in which children are reared, which may have consequences for their long-term chances of success (McLanahan and Sandefur, 1994; Case et al., 2002; Bertrand and Pan, 2013).

The analysis in this paper focuses on the marriage decision immediately following a non-marital pregnancy, sometimes referred to as a "shotgun" marriage (Akerlof et al., 1996). There are two reasons for focusing on this marriage decision. First, an influential paper by Akerlof et al. (1996) finds that the decline of marriages following an out-of-wedlock pregnancy explains, in an accounting sense, the majority of the increase in out-of-wedlock births between 1965 and 1990. Second, these marriages are likely to be the most elastic with respect to

[1]To provide a benchmark sense of child support's reach and expansion, in the fiscal year 2013 there were 15.6 million child support cases and $31.6 billion of total distributed child support collections, compared to 4.1 million cases and $3.6 billion in the fiscal year 1978, valued at 2013 USD (source: Office of Child Support Enforcement, FY2013 Preliminary Report - Table P-52 and Carmen Solomon-Fears, 2005). The majority of the increase in child support cases and collections has been among children of never-married couples rather than of divorced couples (Freeman and Waldfogel, 2001).

child support laws: a couple with a pregnancy deciding whether to marry will consider what happens if they do not, a state of affairs governed directly by child support. Prior to the child support era, if such a couple decided against marriage the child would have had no legal father. With a shotgun marriage, however, the husband was (and still is) granted the presumption of paternity (Edlund, 2013), which was accompanied by custodial rights and financial obligations to support the child.[2] Hence, unmarried couples facing a pregnancy face a direct and immediate tradeoff between marriage, single motherhood, and aborting the pregnancy. I explore these mechanisms with a theoretical model that illustrates how child support may crowd out the commitment role of marriage.[3]

In the empirical analysis of this paper, I first show that child support laws at the time of pregnancy have an effect on child support income receipt in subsequent years, for mothers who do not marry. Using variation in the timing of pregnancy relative to the rollout of child support legislative enactments, I show that the full rollout of child support laws can account for around a 5 percentage point increase in the probability of a never-married mother receiving any child support income, off a base of 17 percent. The mean dollar amount of child support received, conditional on receipt, is between $2000 and $3000 per year (expressed in 2000 USD), and represents about 10-15 percent of annual household income for these families. This evidence is consistent with previous literature finding an effect of child support laws on the mother's income.[4]

Next, I consider the effect of child support laws on the marriage decision following a nonmarital pregnancy. The empirical design is motivated by the idea that the exact timing of conception relative to the child support legislative rollout is difficult for the individual to

[2]A marriage *after* the birth of the child would have established the father as a step-father only, and if he desired to establish legal parental rights he would have had to go through legal proceedings to formally adopt the child.

[3]Evaluating the welfare consequences of a child support policy for children requires an understanding of both the potential negative effects on family structure and the increased resources to single-parent households, an analysis that is beyond the scope of this paper.

[4]See, for example, Beller and Graham (1991); Miller et al. (1997); Argys et al. (2001); Freeman and Waldfogel (2001); Sorensen and Hill (2004).

control. To consider the effect of child support on the marriage decision, I use variation in the spatial and temporal rollout of state child support laws relative to the timing of nonmarital pregnancy. The main specification assumes that propensity to marry evolves in the same way for women who are pregnant as it does for those who are not. This assumption is later relaxed by allowing first a linear time trend and then separate year interactions with pregnancy. I perform a variety of specification checks to demonstrate that the estimates are robust and are not driven by a violation of common trends.[5] All specifications support the interpretation that child support has reduced the likelihood of couples marrying following a nonmarital pregnancy.

An important challenge for identification in this setting is selection: the main regression is estimated under the assumption that child support policies are not affecting female fertility differentially based on her underlying propensity to marry. It is plausible, for example, that child support discourages fertility among women who are most likely to marry. To explore this possibility, I estimate the direct effect of child support on fertility and find the estimates negative, but small and insignificant, too small by an order of magnitude to explain the estimated effects on shotgun marriages.[6] I also perform pessimistic counterfactual exercises that assign higher and higher marriage rates to non-pregnancies, and re-estimate the main specification.

[5]First, I show that baseline characteristics, including both outcome variables and controls, do not predict future innovations in child support laws, reducing the concern for reverse causality. Second, I control directly for societal attitudes at the region level, which proxy for public sentiment on issues including premarital sex, sex education, abortion, and the generosity of public assistance. Third, I show that the estimates are robust to specifications allowing for differential trends related to baseline state-level characteristics. Fourth, I allow the propensity to marry to vary flexibly with time separately by pregnancy state; the main effect remains similar in sign and magnitude but loses statistical significance. An F-test of the null hypothesis that the interactions between year dummies and pregnancy state are jointly equal zero fails to reject, lending support to the main specification and mitigating concern that differential trends in the propensity to marry by pregnancy state are driving the results.

[6]Aizer and McLanahan (2006) find a negative effect on total fertility for those with a high school degree or less, and slightly positive effects for those with some college or more. This finding works against finding a negative effect on shotgun marriages, however, since low-educated women are less likely to have a shotgun marriage than high-educated women. The theoretical model presented in Section 3 supports their finding: those who are least likely to marry following a pregnancy reduce their fertility the most. Both pieces of evidence suggest that the estimates presented in this paper are a lower bound, in absolute value, of the true effect.

The regression estimates suggest that the full set of child support laws adopted by U.S. states over this period reduced the probability of a marriage following a nonmarital pregnancy by about 7.7 percentage points, off a base of 38 percent. This number is quite large when compared to the total decline in the fraction of nonmarital pregnancies resolved in a marriage before birth, about 10 percentage points over the 1977-1992 sample period for which the child support legislative data is available. These "shotgun" marriages represent about 10 percent of all first marriages, and hence this effect is a relatively minor change in marital patterns overall. However, the effect is large enough to have a substantial impact on nonmarital birth rates; a back of the envelope calculation suggests that a decline in shotgun marriages of this magnitude can account for the fraction of nonmarital births increasing by about 2.7 percentage points, relative to its total increase of 11 percent over the sample period.[7]

Next I estimate the effect on the state-level abortion rate, which is defined as the number of abortions per 1000 women aged 15-44, in a state-year. Child support may affect the abortion rate by lowering the cost of raising a child as a single mom. I show that the full set of child support laws reduced the abortion rate by 1 to 2 per 1000 women, relative to a base of 28. These results again rely on the somewhat strong assumption of common trends, which will be violated, for example, if state-level trends in attitudes toward abortion and child support laws are correlated or jointly determined. For this reason I control directly for attitudes toward the legalization of abortion, and the composition of state legislatures and governors' offices. In addition, I subject the estimates to the same battery of specification checks and robustness exercises as described above for the marriage estimates. The results hold up well.

This paper contributes to the literature studying long-term marital trends, including the decline of shotgun marriages (Akerlof et al., 1996; Alesina and Giuliano, 2006) and

[7]This calculation follows Akerlof et al. (1996) and can be made as follows. Define the *shotgun marriage rate* as the fraction of couples who marry between pregnancy and birth. Then the change in the fraction of children born outside of marriage due to a change in the shotgun marriage rate, holding the fraction conceived outside of marriage fixed, is: $(1 - s_{t+1})b_{t+1} - (1 - s_t)b_{t+1}$, where s_t represents the shotgun marriage rate at time t and b_{t+1} represents the fraction conceived outside of marriage at time $t + 1$.

the rise of nonmarital births (Willis, 1999). Akerlof et al. (1996) documents the decline in marriage following nonmarital pregnancies between 1965 and 1990, and shows that this decline accounts for the majority of the increase in births to unmarried women over this period. Akerlof et al. (1996) presents a theory in which the availability of contraceptives and abortion can reduce shotgun marriages. This paper is complementary to their work, presenting both theoretical support and empirical evidence that child support is a key factor behind the dramatic decline of these marriages.[8]

Several studies find that child support laws have led to a decline in total nonmarital births (Case, 1998; Plotnick et al., 2007; Aizer and McLanahan, 2006); Rossin-Slater (2013), on the other hand, finds that the adoption of in-hospital paternity establishment programs reduces post-birth parental marriage, and hence *increases* nonmarital parentage. The theoretical model of this paper rationalizes these findings. I show that a decline in total nonmarital births can occur because men reduce fertility to avoid costly unwanted births. However, conditional on a pregnancy, marriage rates decline and the *fraction* of births that are born outside marriage increases. The empirical evidence presented in this paper is consistent with this unified interpretation.

This paper also contributes to the literature studying determinants of the abortion decision. Prior research has considered the role of the welfare system (Moffitt, 1998), Medicaid funding (Henshaw et al., 2009) and legal restrictions on providers (Haas-Wilson, 1996; Joyce and Kaestner, 2000). This is the first paper to my knowledge to consider the effect of child support laws on abortion rates. Moreover, there is little work in economics attempting to understand the causes behind the decades-long decline in the abortion rate, which peaked in the early 1980s at 29.3 abortions per 1,000 women aged 15-44 years but has subsequently declined to 19.6 as of 2008 (Jones and Kooistra, 2011). This paper provides evidence that child support is an important factor.

[8] An important distinction between the Akerlof et al. (1996) model and the one presented in this paper is that Akerlof et al. (1996) present the decline of shotgun marriages as welfare-diminishing for marginal women who no longer marry, while here it is *welfare-enhancing* for these women, because they receive additional transfers outside marriage.

The rest of the paper is organized as follows. Section 2 presents a brief overview of the child support process and describes the legislative data used in the empirical analysis; Section 3 presents a conceptual framework to explore mechanisms and consider welfare consequences; Section 4 describes the data and presents summary statistics and empirical trends. Section 5 presents the identification strategy, the main results, and robustness exercises; and Section 6 concludes.

2 Background on child support

The first major federal policy affecting child support was the creation in 1975 of part D of Title IV of the Social Security Act, which created the federal Office of Child Support Enforcement (OCSE).[9] The primary purpose cited in these laws is twofold: to promote the best interests of the child by promoting financial stability within families, and to decrease dependence on public assistance.

The child support process

At the individual level, the child support process begins with paternity establishment. If the mother is married at the time of birth, the husband is presumed to be the father. If the mother is unmarried at the time of birth, the state or child support agency will attempt to determine paternity, according to the state laws in effect. There have been significant changes in the procedure for establishing paternity since the 1970s. In 1978, 111 thousand paternities were established, increasing to 1.6 million in 1999 (U.S. House U.S. House of Representatives, 2014).[10]

[9] Lerman and Sorensen (2001) provides a useful overview of child support enforcement. See also (U.S. House of Representatives, 2012), which has a concise legislative history of Federal laws regarding child support.

[10] Federal laws began imposing minimum paternity establishment rates upon states in 1988: initially set at 50 percent, it has been incrementally increased to 90 percent, as mandated by the 1996 welfare reform law. A provision of the Family Support Act of 1988 requires states to pay the up-front cost of genetic testing, and to mandate a genetic test in a contested paternity case at the request of any party. Most states now have in-hospital Voluntary Acknowledgement of Paternity programs, which present acknowledgement forms

If paternity is not established or presumed, a biological father has no rights or obligations to his child. From a legal perspective, the child and biological father are strangers. For centuries, governments and church parishes have attempted to reduce their welfare rolls by locating absent fathers and getting them to financially contribute – a notable early example is the 1576 Poor Law of England. In practice, paternity establishment has been revolutionized by late-20th century developments in technologies that assist in identifying, or ruling out, a potential father, such as blood typing and genetic testing (U.S. Department of Health and Human Services, 2002).[11]

Once paternity is established, the next stage of the child support process is to obtain a child support order. The process for obtaining a child support order has been transformed and streamlined since 1975. If the mother is a recipient of public aid (IV-D cases), the IV-D agency will initiate child support proceedings against the person alleged to be the father, beginning with paternity establishment and through securing a child support order. A mother who is not a recipient of public aid has to bring a child support action to court herself, beginning with a paternity action and followed by a petition for child support. The child support order itself specifies the monthly child support amount, after consideration of the parents' income sources and the state guidelines for child support payments. The child support order will also specify the method of payment, which now may include wage garnishment or automatic withholding from public assistance such as unemployment benefits, Social Security Income, or tax returns.

to the man accompanying the mother in the hospital at the time of birth (Rossin-Slater, 2013).

[11]For divorced couples – that is, cases in which paternity is not legally in doubt – child support orders have been a component of alimony since before the 1975 federal law, although data on the frequency of payments or compliance with these orders prior to the creation of the OCSE are difficult to find.

3 Model

This section presents a simple model to explore both the mechanisms and the welfare consequences of a child support policy for the mother and father.[12] Some predictions are straightforward; for instance, that women who would be single moms without child support are better off with a child support transfer. Other predictions are more subtle: for example, that women who would have a child and marry regardless of child support and hence never use it, are better off; or that some men benefit from child support, because it creates an ability for them to commit to future transfers which they could not credibly do before. Unlike prior models of child support (Aizer and McLanahan, 2006; Rossin-Slater, 2013), the framework here endogenizes both marriage and fertility decisions, making it better suited to study the effect of child support on *total* nonmarital fertility separately from the *fraction* of births that are born outside of marriage. The model also introduces abortion as a third alternative to marriage and single motherhood.

Model setup

There exists a continuum of men and women who derive utility from one private composite good c, from sex, and from children.[13] Individuals live for two periods, and discount the future by a factor β.

Men and women are heterogeneous in their preference for children. Let v_F denote a woman's individual-specific preference for children, assumed to be distributed over a support $[-V, V]$ with distribution G_F, and let v_M denote a male's preference for children, distributed over the same support with distribution G_M. Taste parameters for children are assumed to be perfectly observable to both partners.

Information is perfect: all parameter realizations, including one's own value of children

[12] I leave aside the equally important consideration of the welfare of the child, which is beyond the scope of this paper but is discussed in Rossin-Slater (2013).

[13] For simplicity, we consider the choice between having one child and none, although the framework easily generalizes to the case of multiple children.

and one's partner's value of children, are perfectly known, and are known at the start of the game.

In period 1, all men and women are single and childless, and are matched at random into couples. Men and women receive period 1 utility from sex only, which generates a fixed surplus $b > 0$ for each partner. Period 1 utility is simply $U_i = b \cdot s_i$, for $i \in \{M, F\}$, where s_i is an indicator for having premarital sex; individuals receive no wages and have no private consumption in period 1.

At the beginning of period 2, couples find out if there is a pregnancy, which occurs with exogenous probability λ if the couple had premarital sex in period 1. Women can abort the pregnancy at individual-specific cost δ distributed over support $\mathbb{R}^+$, which encapsulates any psychic, distance, or monetary costs, and is perfectly observable to both partners.

Men and women receive period 2 utility from the private consumption good and from children. In this simple setup, women are assumed to be the default custodian of their children, and by assumption they derive equal utility from children whether or not they are married. Female utility in period 2 is written $U_F(c_F, k, a) = u(c_F) + v_F \cdot k(1-a) - \delta \cdot a$, where c_F denotes consumption of the private good, normalized to have unit price, k is a dummy variable indicating a pregnancy, a is a variable indicating if the pregnancy is aborted, and $u(\cdot)$ is assumed to be strictly increasing and concave.

Men receive v_M from children if they are married, which is diminished by a factor α if they are not married, where $0 < \alpha < 1$. This assumption reflects the lessened custodial rights of fathers outside of marriage or a father's emotional distance from living away from his child (Chiappori and Oreffice, 2008; Edlund, 2013; Browning et al., 2014). Period 2 male utility in marriage is $U_M(c_M, k, a) = u(c_M) + v_M \cdot k(1-a)$, and outside marriage is $U_M(c_M, k, a) = u(c_M) + \alpha v_M \cdot k(1-a)$. I emphasize that c_M and c_F are not constrained to be the same inside and outside marriage.

In period 2, men earn wages w_h. Women earn w_h too, unless they have a child, in which case they earn $w_\ell < w_h$, reflecting both diminished earnings due to time away from work

and the cost of raising the child (Bertrand et al., 2010). Women who choose to abort the pregnancy keep their full wages in period 2.

Timing of marriage and fertility decisions

In period 1, the couple decides on premarital sex. If both partners agree to premarital sex, then with probability λ the female partner becomes pregnant at the beginning of period 2. (The game ends with outside options if the couple decides not to have sex.)

In period 2, if there is no pregnancy there is no surplus of marriage and the game ends with outside options. If there is a pregnancy, the couple decides on marriage; if both partners agree, they marry and bargain over marriage allocations; otherwise they remain single.[14] If the couple remains single, the female decides whether to abort the pregnancy or not.[15] Note that bargaining over marriage allocations occurs at a time when the female partner considers abortion as one of her outside options.

Married couples bargain to a Pareto efficient outcome: if both partners cannot be made better off under marriage they do not marry. If both partners can be made better off under marriage, the couple marries and consumption is allocated to maximize weighted utility subject to both partners being better off, with a weight of 1 on the female partner and μ on men (Browning et al., 2014).

The model is solved by backwards induction in Appendix A.

[14]The timing of marriage and fertility decisions in the model is such that all marriages are "shotgun marriages," in the sense that they are triggered by the pregnancy and would not occur otherwise. One could include an additional match-specific non-monetary benefit of marriage θ, as in Chiappori et al. (2006), which would introduce both shotgun and non-shotgun marriages.

[15]Because children are the only surplus of marriage and preferences for children are public, no abortions will occur in marriage.

Child support

The main focus here is the impact of a *child support policy*: a government-enforced lump sum transfer $\tau > 0$ from the father to the mother in the case of nonmarital fertility.[16] I impose one further assumption about the size of τ: that τ is small enough so that at least some men with distaste for children – i.e. some men with $v_M < 0$ – are still willing to have premarital sex.

The main implications of introducing a child support policy are summarized with the following proposition.

Proposition 1. If a child support policy is introduced, the following occur:

1. Strictly fewer women have an abortion, conditional on pregnancy.

2. Women who marry following a nonmarital pregnancy enjoy weakly more private consumption.

4. The effect on total pregnancies is ambiguous.

3. A set of couples unable to commit to marriage will have premarital sex.

Proof. See Appendix A. □

The intuition behind Proposition 1 is as follows. Under a child support policy, strictly fewer women abort a pregnancy because being a single mom is more attractive with added financial support from the father. Married women have greater bargaining power in their marriage because their outside option – being a single mom – is now more attractive, raising the private consumption required for them to marry. The total effect on pregnancies is ambiguous because, while women's willingness to become pregnant increases, men who dislike children are less willing to have premarital sex. However, a set of couples that previously faced a commitment problem will now have premarital sex and will not marry. Prior to child support a set of men were willing to commit *ex ante* to future transfers to have premarital sex,

[16] An important feature of child support is to provide additional custodial rights to the father, which would be represented here by an increase in α for non-married fathers; I consider this case in Appendix A.

but this commitment was not credible; *ex post* these man would walk away. The commitment problem is solved by child support, which enforces commitment from men in couples who will not marry.

Welfare

All women are at least weakly better off by the introduction of child support. Women who marry have weakly higher bargaining power within marriage. Marginal women who prior to child support would have sex only with commitment and now will have sex without commitment to marry are better off too, because they receive higher consumption in marriage and receive the utility benefit of sex. Infra-marginal women who, prior to child support, would have sex without commitment are strictly better off, too, because they now receive child support payments in the event of a nonmarital pregnancy. Women who continue to reject premarital sex are no worse off than they were prior to the reform.

Let us now consider men. Men who can credibly commit to marriage ($v_M > 0$) – are weakly worse off, because they have weakly lower bargaining power in marriage. Some men are weakly better off: those for whom child support solves the limited commitment problem. These men are better off because without child support they were willing to commit *ex ante* but *ex post* they would walk away; child support enforces commitment from these men.

It is important to emphasize that in this model, child support is welfare-enhancing for the marginal women who no longer marry. This is in contrast to the shotgun marriage model of Akerlof et al. (1996), in which a decline in shotgun marriages is welfare-diminishing for these marginal women.[17]

[17]The decline in shotgun marriages in the Akerlof et al. (1996) model arises as a result of a new technology – contraceptives and abortion – which is adopted by an exogenous fraction of women. The adoption of this technology by a large enough fraction of women induces the shotgun marriage equilibrium to switch to a second equilibrium in which men no longer have to promise to marry as a means for obtaining premarital sex. The exogenous fraction of women who fail to adopt the new technology are worse off because they can no longer extract a shotgun marriage promise, and no longer marry. In the model presented here, by contrast, the marginal women are better off because they get additional transfers by not marrying their male partners.

4 Data sources

This section describes the data sources and presents descriptive statistics.

4.1 Child support legislative data

This study uses a state-year database of child support legal enactments, used in prior work by Case (1998) and Plotnick et al. (2007), and which includes laws at each stage of the child support process.

This data is available for the years 1977-1992 and for all U.S. states with the exception of Hawaii and Alaska. I construct a state-year legislative index, denoted CS_{st}, which is an average of 9 indicator variables, each of which represents a different child support law and takes a value of 1 if the associated law is in effect in state-year st. Hence the index is a continuous variable ranging from 0 to 1, where 0 reflects that the state has none of the 9 laws in effect and 1 reflects that the state has all 9 laws in effect. A single-dimensional index increases power in the analysis and allows us to draw inferences about the aggregate impact of the rollout, but it has the disadvantage of imposing linearity so that each law is assumed to have the same marginal effect, abstracting away from any heterogeneity in their impact. I explore heterogeneity in the main results in Appendix B.

The 9 laws included in the analysis are: a law requiring immediate income withholding for new or modified child support cases, deducting child support obligations from the obligor's paycheck; a law allowing the custodial parent to place a lien on the non-custodial parent's property; a law permitting genetic tests to be used to resolve disputed cases; a law allowing paternity to be established while the child is below 18 years of age; a law requiring the provision of local child support collection and enforcement services to non-recipients of AFDC; a law requiring withholding of arrearages from parents who are delinquent on their payment; a law creating criminal penalties for failure to pay child support; a law establishing the right to bring a child support action against a parent residing in another state; a law

creating a central registry for child support payments.

Figure 1 plots the mean adoption rate across states of each of the 9 different child support statutes used in the analysis. The figure shows a dramatic increase in adoption of statutes, and considerable variation in the timing and abruptness of enactments across states. While incremental changes to child support collection and enforcement have taken place since the 1970s, with the gradual trend being toward greater enforcement and stronger measures to promote collection, certain federal laws brought abrupt changes, which can account for the dramatic rise in certain types of laws. For example, in 1984, Congress enacted Public Law 98-378, which required states to implement income withholding procedures for parents delinquent on payments, and hence in Figure 1 we see a big jump in state laws withholding for delinquency between 1984 and 1986. Similarly, Public Law 100-485 was enacted in October of 1988 and implemented income withholding procedures that would take effect immediately, unless the parties were able to come to an alternate arrangement; Figure 1 shows a spike in state laws introducing immediate withholding between 1988 and 1990.[18]

4.2 Child support income data

Child support income data comes from the IPUMS March CPS (Flood et al., 2017).[19] Figure 2 shows that child support income increased substantially over this period.[20] The series plotted on the left y-axis shows the fraction of never-married mothers with a custodial child

[18] Accompanying the strengthening of policies, child support enforcement expenditures per single mother roughly tripled between 1980 and 2000, from $200 per single mother family to approximately $600, in constant 2000 USD (Freeman and Waldfogel, 2001).

[19] The March CPS has an advantage over alternative data sources recording information about child support in that it offers the longest time series, allowing one to observe if a mother received any child support going back to 1977. By contrast, the April CPS Child Support Supplement was introduced in 1979 and has been repeated every other year since 1982. SIPP, which also records information about child support receipt, has its earliest sample in 1984. The March CPS has the disadvantage in that it does not contain information on child support awards or amount owed by the non-resident parent.

[20] For years 1977 to 1987 I use the GOTALCH variable, which is an indicator for the respondent having received alimony or child support income over the past year. I use the sample of never married women only, hence a positive response indicates child support income not alimony. The March CPS began collecting data on child support as a separate income category in 1988, and I use this CSINCOME variable for 1988 and later years.

under 18 and an absent father who report receiving any child support income, for years 1976-2010. The series plotted on the right y-axis shows the average annual pre-tax child support income in 2000 USD for those reporting any child support income. The second series shows that the dollar amount received per mother has been steadily increasing over the sample period, and the amount represents between 10 and 15 percent of total household income for mothers in the sample. (I omit the period 1977-1987 from this second series because it is not possible to separate child support income from other residual income categories in this period.)

Note that in spite of the dramatic increase in the fraction of never-married mothers receiving child support from 1980-2000, the fraction never surpasses 25 percent. This may reflect underreporting of income, which has been documented in the March CPS for child support and other transfer programs, and appears to be more pronounced among lower income respondents (Wheaton, 1997; Meyer et al., 2009; Meyer and Mittag, 2014).[21]

4.3 Marriage and fertility data

The marriage and fertility data used in the main analysis comes from the restricted Survey of Income and Program Participation (SIPP), which is available in the Census Bureau's Research Data Center. I use years 1992, 1993 and 1996, which have retrospective marital and fertility histories in Topic Module 2 as well as state identifiers and individual demographic controls. The state of residence identifier in SIPP allows me to link this data to the state-level child support legislative data.[22]

I emphasize that the restricted SIPP does not include data on miscarriages or abortions; hence the analysis of marriage following a first pregnancy is, more precisely, an analysis of marriage following a pregnancy that results in the first live birth reported by the respon-

[21] In terms of dollar amounts, according to the OCSE's 2000 annual report, the amount of current support due in FY 2000 was $23.03 billion, while the amount of current support distributed was $12.9 billion, or 56 percent of the total.

[22] Note that all reported numbers of observations and degrees of freedom in F-statistics that come from the restricted SIPP data are rounded according to Census disclosure requirements.

dent.[23]

Table 1 displays sample means and standard deviations for the SIPP sample. The sample includes women aged 18-44 in each time period given in the column heading. I define a "shotgun" marriage as a first marriage occurring between 0 months and 8 months prior to the first birth, following Akerlof et al. (1996). Figure 3 shows a three-year moving average of the *shotgun marriage rate*: the fraction of year t nonmarital (first) pregnancies resolved in a marriage 0 to 8 months prior to birth.[24] In 1970, the shotgun marriage rate is over 50 percent and declines to slightly over 20 percent in 1996.[25] It is important to stress that the decline in shotgun marriages preceded the strengthening of child support laws by about 5 years, hence the decline cannot be explained completely by the advent of child support laws.[26] Figure 3 also plots the fraction of total first marriages in year t that are shotgun marriages, showing a broadly stable fraction until the early 1990s, and then declining.[27]

4.4 Abortion data

State-year data on abortion totals and the number of abortion providers comes from the Guttmacher Institute and is based on periodic surveys of abortion providers (Henshaw and

[23]For conciseness, throughout the empirical analysis using SIPP data, I use "pregnancies" to refer to pregnancies that result in a birth. Section 5.3 considers the direct effect of child support on abortions using aggregated data at the state-year level. Section 5.4 considers the potential for selection bias in the marriage analysis that may occur due to missing data on miscarriages and abortion.

[24]Because only live births are observed, the shotgun marriage rate is computed as the fraction of women who are never-married as of 9 months prior to their first birth who become married between 0 and 8 months prior to the birth.

[25]The decline in shotgun marriages presented here is consistent with Akerlof et al. (1996) and complements the empirical finding in the sociology literature that since the 1970s, pregnancies outside marriage are far more likely to be resolved in cohabitation rather than marriage (Lichter et al., 2014).

[26]Child support was introduced in part as a response to the increase in the number of single-mother families on welfare. While child support may have succeeded in reducing welfare receipt among single moms (Huang et al., 2004), this paper argues that it may have had the unintended consequence of actually *increasing* the fraction of single-mother families, through its effect on the marriage decision.

[27]Appendix D presents additional descriptive analysis of shotgun marriages, based on the National Survey of Families and Households and SIPP, showing that they differ qualitatively from non-shotgun marriages in terms of marital happiness and divorce hazard rates. Shotgun marriages are less likely to report being happy in their marriage, more likely to report having discussed separation with their spouse, and more likely to divorce.

Kost, 2008). The data reports total abortions and abortions per 1000 women aged 15-44, both by state of residence and state of occurrence, and is available for years 1978-2008, with gaps in years 1983, 1986, 1989, and 1990 in the analysis period. I use abortions by state of residence rather than state of occurrence since child support laws will affect women by their state of residence. The data also includes counts of the number of abortion providers in each state and year; these include hospitals, clinics, or physicians' offices where abortions are performed. I use these counts to construct a proxy for abortion availability: the number of abortion providers per 100,000 women aged 15-44.

Figure 4 depicts trends in total abortions and in the abortion rate, which is defined as the number of abortions per 1000 women aged 15-44. The decline in the abortion rate over 1978-1992 continues until about 2004, although the child support legislative database does not cover such recent years. Table 2 presents descriptive statistics of the abortion data over the sample period used in the analysis. We see that total abortions declined from 28.7 per 1000 women per year in the 1978-1980 period, to 25.9 in the 1990-1992 period. This downward trend has been accompanied by a downward trend in the number of abortion providers per 100,000 women aged 15-44, from 5.3 in the 1977-1980 period to 4.1 in the 1990-1992 period.

4.5 State-year controls, societal attitudes and political affiliation data

The state-level demographic controls come from the March CPS. Table 1 presents summary statistics of the mean and standard deviations of state-level controls used in the main analysis, including percent black, male unemployment rate, share of college degree holders who are female, the female labor force participation rate, share of the state population living in an urban area, share of state residents collecting welfare, share of single moms on welfare, and the poverty rate.

Survey data on attitudes toward abortion, sex, marriage, welfare, and women come from the General Social Survey (GSS). I include variables from the GSS as controls in the analysis

to lend further support that trends in attitudes are not behind the main results on marriage and abortion. See Appendix Table B.1 for further details on the survey questions used with their full text, and the years available. I restrict the data to survey respondents aged 18-44, since this is the age group considered in the fertility and marriage analysis; I aggregate survey responses to the region-year level.[28]

I use data on the majority party of state legislatures, and the party affiliation of governors, which was assembled by Halvorsen and Jakobsen (2013). The data on state legislatures comes from the National Conference of State Legislatures (NCSL), while data on state governors are from the National Governors Association (NGA). Table 2 summarizes the eight possible combinations of party leadership (Democratic or Republican) of the State House, State Senate, and the governor's office.[29]

5 Empirical strategy

This section first explores determinants of state-level adoption of child support statutes, presenting evidence that state-level characteristics are not predictive of the timing and geography of statute adoption. I present evidence that these laws indeed have had an effect on the likelihood of receiving child support income. I then consider the effect of child support laws on state-level abortion rates. Lastly, I present the main estimating equation for child support's effect on marriage and discuss the assumptions required for identification, and present the results.

[28]State identifiers are not available in the public-use GSS data. Even with access to state identifiers, however, sample sizes become too small: there are approximately 1700 total responses per year over the 1977-1992 period.

[29]The "other" category refers to states with independent governors. Nebraska has a unicameral legislature, and its data is coded as having the same party affiliation for both House and Senate.

5.1 The child support rollout

The empirical strategy of this paper is motivated by the idea that the exact timing of pregnancy relative to the legislative regime is difficult for the individual to control. Identification in our setting nevertheless faces the challenge that state-level trends in supply and demand factors for marriage may be correlated with the timing of states' adoption of child support statutes, which I explore now.

As a preliminary step, I regress the child support enforcement index defined above on state fixed effects, time fixed effects, and time-varying controls that are likely to covary with marital attitudes or marital opportunities. I find that 91.7 percent of the variation in the child support index can be attributed to time-invariant state effects and common time effects, while 0.03 percent of the variation in the child support index can be attributed to time-varying factors including: welfare recipiency rate, male unemployment rate, single mom share on welfare, female participation rate, female college share, percent black, urban share, poverty rate, college-high school premium (reported in Appendix Table B.2).

Next, I consider whether baseline state characteristics predict future innovations in child support laws. The idea here is that state-level variables, including both outcomes of interest and controls, are correlated with supply and demand factors for marriage. If they predict future innovations in the law it would suggest either (1) a reverse causality story, in which changes in marriage or fertility behavior are drive child support law adoption; or (2) a violation of the common trends assumption, because baseline state characteristics are predictive of the future path of their adoption of laws.

I estimate the following regression:

$$\Delta CS_{st} = \alpha + \psi_t^{'} w_{s,1977} + \eta_s + \xi_t + \epsilon_{st} \tag{5.1}$$

where $\Delta CS_{st} = CS_{st} - CS_{s,t-1}$, and $w_{s,1977}$ is a vector of baseline state-level outcome variables and controls, normalized to have mean zero and standard deviation 1 in the baseline

year. The vector $w_{s,1977}$ includes the following: the shotgun marriage rate, the abortion rate, the 10 year divorce hazard, the black incarceration rate, the fraction black in the population, the male unemployment rate, the share of college graduates who are female, the female participation rate, the share of urban residents, the fraction of the population receiving welfare, the fraction of single moms on welfare, the poverty rate, and the college-high school premium. The baseline year is 1977 for all variables in $w_{s,1977}$, except, because of data limitations, the black incarceration rate, which has the baseline year 1980, and the abortion rate, which has the baseline year 1978. The regression includes state fixed effects and year fixed effects (η_s and ξ_t, respectively).

The estimates $\hat{\psi}_t$ can be interpreted as the effect of increasing by 1 standard deviation a state's baseline level of a particular characteristic on a future increase in the child support law index. Figure 5 plots the estimated coefficients $\hat{\psi}_t$ and their associated 95 percent confidence intervals. The first row of Figure 5 shows that baseline levels of the main outcome variables studied in this paper – the shotgun marriage rate and the abortion rate – do not predict future innovations in child support laws and do not appear to be driving changes in child support statute adoption.

The second and third rows of Figure 5 show that baseline state-level controls do not appear to predict future innovations nor time trends in the child support law index. Female marital opportunities, which may be related to the female labor force participation rate, the female share of college degree holders, or the fraction of African-American men incarcerated, do not appear to predict future changes in child support statute adoption.[30] Taken as a whole, the panels of Figure 5 suggest that the timing of the rollout has no clear systematic relationship with baseline state-level marital patterns or demographics. Appendix C explores this question further using the social attitudes data from GSS, finding that social attitudes

[30]Certain variables, such as the share receiving welfare, the poverty rate, and the share of single moms receiving welfare are borderline statistically significant in years 1986-87. To formally test significance, I perform an F-test of each variable's year coefficients being jointly equal to zero and report the results in Appendix Table B.3. In none of the cases do we reject the F-test of equality at the 10 percent significance level. I also fail to reject an F-test of all variables' year coefficients being jointly equal to zero.

do not correlate strongly with law adoption once we remove region and year fixed effects.

5.2 The effect of child support laws on income

This section aims to show that child support laws at the time of pregnancy indeed have an effect on future child support income for single moms.

I restrict the analysis to the population of never-married mothers who report having a custodial child in the household under 18 years of age, and who also report the father living outside the household. I then estimate the following regression:

$$Inc_{ist} = \beta_0 + \beta_1 CS_{s,t-a-1} + \gamma' x_{ist} + \alpha' w_{st} + \epsilon_{ist} \tag{5.2}$$

where Inc_{ist} is an indicator taking a value of 1 if person i receives any child support income in year t, $CS_{s,t-a-1}$ represents the child support law index at time $t-a-1$, where t is the survey year and a is the age of her youngest child, and s is the state of residence.[31] The state of residence at the time of the survey is assumed to be the state of residence at the time of the youngest child's birth (an assumption explored in Appendix B). The vector x_{ist} represents individual controls, including age, age-squared, age at the time of first birth, race indicators, dummies for the four education categories, number of children dummies; w_{st} is a vector of state-year controls; ϵ_{ist} is an error term representing the unobserved determinants of child support receipt. Throughout the paper, standard errors are always clustered at the state level and are robust to heteroskedasticity.

In this regression we expect the sign of β_1 to be positive, indicating that the adoption of a child support law has a positive effect on the fraction of women reporting child support income receipt. Table 3 indicates that this is indeed the case. Since the child support index is constructed with 9 laws, we can interpret the coefficient as representing the effect of moving from a regime with none of the laws in place to one with all 9 laws, on the

[31] I use the age of the youngest child because child support laws have strengthened over time, and these laws are most likely to impact child support receipt in the case of a mother's most recent child.

probability of receiving child support income. Here these laws increase the probability of child support receipt by 5.3 percent, in the specification with state and year fixed effects (column 3). Adding additional time-varying state controls (column 4) and baseline (1977) state-level controls interacted with time trend (column 5) do not reduce the magnitude of the laws' effect. Given that the fraction of the sample reporting child support income is 17 percent, the magnitude of the laws' cumulative effect is quite large.[32]

We might be concerned that states that adopt stronger child support laws have a population of individuals more likely to comply with such laws; in this case β_1 may reflect state-year specific attitudes toward the legal system or child support laws rather than the effect of the laws *per se.* Because there is variation within state-year cells in the age of the youngest child it is possible to include interacted state-year fixed effects, which controls for state-level time trends in attitudes toward child support that might lead to greater compliance. The identifying variation in this specification is variation in laws *at the time of birth* of the youngest child, within a state-year cell, and the estimated effect is quite similar, reported in column 6 of Table 3. Other coefficients may be of interest as well: from Table 3 we see that older, higher educated, and non-minority mothers are more likely to report having received child support income.

How much income did the laws generate for single-mom households? We saw in Figure 2 that conditional on receiving any child support income, the average amount received per never-maried mother is between $2000 and $3000, valued in 2000 USD. However, there may be some crowd-out in public assistance income due to policies limiting the pass-through of child support for women receiving public benefits (Cancian and Meyer, 2005). In Appendix Table B.5, I explore the crowd-out of public assistance and SSI receipt for the March CPS sample of never-married mothers. I find that the child support rollout reduced average per-person public assistance from the government by $250-500 annually (also in 2000 USD).

[32]There is also a possibility of measurement error in the CS variable, which may attenuate the effect of the policy on income.

Hence the net benefit of child support is still large as a percentage of income.[33]

5.3 Abortion

In this section I consider the effect of the child support rollout on the abortion rate. The mechanism highlighted in the model is that child support reduces the financial burden of raising a child as a single mom, and women will abort fewer pregnancies as a result. Note that abortion became legal in all U.S. states several years before the child support rollout, following the 1973 Supreme Court decision Roe v. Wade.

The dependent variable used in the regression analysis that follows is the state-year abortion rate, which is defined as the total number of abortions per 1000 women aged 15-44.[34] I estimate the following regression:

$$ab_rate_{st} = \beta_0 + \beta_1 CS_{st} + \gamma' w_{st} + \epsilon_{st} \tag{5.3}$$

All specifications include both state and year fixed effects. Cells are weighted by the state's population of 15-44 year old women. A potential source of bias in estimating Equation 5.3 is that trends in state-level attitudes toward abortion may be correlated with the pattern of the child support rollout. To mitigate this concern, I include a proxy for societal attitudes toward abortion using data from the GSS, which is the principal factor from a factor analysis of the 9 abortion-related questions in the GSS, and is aggregated to the region-year level.[35] Because of gap years in both the Guttmacher and GSS data, and to prevent the regressions from dropping observations due to missing data, I impute values for gap years using the

[33] About 15 percent of child support recipients receive Aid to Families with Dependent Children (AFDC) or Temporary Assistance for Needy Families (TANF), and this fraction has remained relatively stable over the sample period (Cancian and Meyer, 2005). Note that crowd-out of public assistance will only take place during the time the mother is eligible for and receiving AFDC/TANF, while single mothers may receive child support throughout the child's first 18 years, and sometimes longer.

[34] Using the log total abortions as the dependent variable produces similar results.

[35] The societal attitudes factor correlates strongly with the number of abortion providers per capita, which can be seen formally in Appendix Table B.4. This correlation with an objective measure of abortion availability suggests that the GSS survey data represents a meaningful proxy for abortion attitudes.

midpoint of the lead and lag values, for each state-year.[36]

Table 4, columns 1-4, presents the main results, adding additional controls in each successive column. The full set of child support laws adopted over the 1978-1992 period is associated with a decline in the abortion rate by 1.8 to 3.1, off a base of 27.6. The preferred specification is column 4, which includes controls for abortion attitudes, state-year controls, and baseline state controls interacted with a linear time trend. Over the sample period 1978-1992, the mean policy index increased by .637. Hence we scale the point estimate (which represents a change from 0 to 1 in the policy index) by the actual growth in the policy index over this period to get an effect size that ranges from 1.15 to 1.97.

The magnitude of the estimates in Columns 1-4 implies that the child support rollout led to a decline in abortions by 4 to 7 percent over the sample period, suggesting that child support legislation has played a significant role in the recent decline of abortions in the U.S. To provide a sense of the magnitude, the abortion rate – i.e. the number of abortions per 1000 women aged 15-44 – declined roughly 2.5 over the 1978-1992 period, meaning that child support can account for over 46 percent of the decline over this period.

We might be concerned that trends in abortion *availability* are somehow correlated with trends in the child support rollout. For this reason, I control for the number of abortion providers per 100,000 women aged 15-44 in the state-year in the column 5 specification; its inclusion does not affect the results.[37]

Table 4, column 6 includes a full set of dummies for the eight possible combinations of Republican v. Democratic control of the State House, State Senate, and the governor's office at the state-year level. While the coefficient is reduced to -1.80 and loses statistical significance, its magnitude still remains sizable and economically important. Moreover, an

[36]The estimates are not sensitive to this imputation, and the results from the estimation of Equation 5.3 with non-imputed data are presented in the Appendix Table B.8.

[37]Including abortion providers in the regression has the advantage of controlling for a proxy measure of cost, as it captures the availability of abortions represented by the density of clinics in a state-year. Its inclusion, however, has the disadvantage of introducing a classic supply-demand endogeneity, because abortion providers may respond to unobserved shocks to demand captured by the error term. Hence for all but column 5 I omit this control.

F-test that the 8 coefficients are jointly equal to 0 fails to reject ($F(7, 47) = 0.94$ with $p = .48$), and hence these controls are omitted from the main specification. Column 7 includes interactions between region and each year separately; this specification allows for heterogeneity across regions in the time path of abortion rates. The standard errors increase as the regression loses power, and the coefficient on the child support index loses statistical significance at the 10 percent level. However, the point estimate changes only slightly, suggesting that a spurious correlation between region-specific trends in abortion rates and the time path of child support law adoption is not driving the main estimates presented in columns 1-4. In addition, an F-test of the null hypothesis that all interactions between region and year dummies are jointly equal to zero fails to reject the null ($F(8, 47) = 1.06$ and $p = .41$)

Bounding selection bias

The results on fertility presented above suggest that child support laws may have reduced the probability of a nonmarital pregnancy. Any pregnancies that are not occurring may have higher or lower abortion rates than those that do occur. We might think, for instance, that child support reduces pregnancy among women most likely to abort, which would lead us to overstate the effect of the policy on abortion rates.

I perform pessimistic counterfactual scenarios to examine the extent of this bias, and report these in Table 5. As in the selection counterfactuals for marriage performed above, I assume that the full rollout of child support laws reduced total pregnancies by 2 percent. I subsequently assign higher and higher abortion rates to these pregnancies, ranging from the 50th percentile to the 90th percentile in the initial year's state-level distribution.[38] Under the

[38]The expression for the imputed abortion rate is: $\hat{ab_rate_{st}} = ab_rate_{st} + (.02 \cdot CS_{st} \cdot births_{s,1978})(ab_rate^{(p)}_{1978} \cdot \frac{1}{births_{s,1978}})$. The first term is the observed abortion rate, i.e. total abortions per 1000 women in state-year st. The second term is the selection term: total pregnancies unobserved due to the policy (proxied with births), multiplied by the number of abortions per pregnancy (again proxied with births), and expressed per 1000 women. The term $ab_rate^{(p)}_{1978}$ indicates the percentile p in the 1978 state-level distribution used to compute the counterfactual.

counterfactual scenario in which these non-occurring pregnancies have the 90th percentile of abortion rates, our estimate loses statistical significance at the 10 percent level. Hence, the negative effect on fertility would have to consist of women with some of the highest abortion rates for our results to disappear.

5.4 The effect of child support laws on the marriage decision

Because SIPP contains retrospective marital and fertility histories, we can construct a panel at the individual-level, which we can use to identify the effect of child support laws on the probability of never-married women becoming married. The unit of observation is a person-year, and I restrict the sample to include women between ages 18 and 44.

To avoid overweighting individuals who appear in the sample a greater number times, I weight each observation by their SIPP person weight multiplied by the reciprocal of the number of total observations for that person in the regression sample, so that each person's observation weights sum to her SIPP person weight. (This reweighting has a relatively minor effect on the estimates, and the unweighted regressions are reported in Appendix Table B.6.)

The main estimating equation is

$$marry1st_{ist} = \beta_0 + \beta_1 nmpreg_{ist} + \beta_2 nmpreg_{ist} \times CS_{st} + \beta_3 CS_{st} + \gamma' x_{ist} + \epsilon_{ist} \qquad (5.4)$$

where $nmpreg_{ist}$ is an indicator for never-married woman i becoming pregnant with her first-born child in year t. The variable $marry1st_{ist}$ is an indicator for person i becoming married for the first time in either year t or year $t+1$. The variable CS_{st} represents the child support enforcement measure for state s in year t.[39] The vector of individual controls x_{ist} includes a high school completion dummy, race dummies, age dummies, and SIPP sample year dummies. I omit higher education categories from the regression controls because

[39]The respondent's state of residence at the time of the survey is assumed to be her state of residence in each year of the panel. This assumption requires there to be no selective migration by women who have a nonmarital pregnancy based on states' child support laws. I check this assumption in Appendix Table B.9 and find no evidence for selective migration.

the decision to get married may affect educational attainment, and their inclusion would introduce a simultaneity bias. All specifications include state fixed effects and year fixed effects, which remove state and year means in the propensity to marry.

The coefficient β_1 in Equation 5.4 represents the effect of having a nonmarital pregnancy in year t on the individual's decision to get married in year t or $t+1$, relative to those who are not pregnant (or who miscarry or have an abortion). The coefficient of interest is β_2, which represents the difference in difference: the effect of strengthened child support enforcement on marriage for women who face a nonmarital pregnancy relative to women who do not, relative to this difference under weakened child support enforcement.

Without the inclusion of interaction terms between $nmpreg_{ist}$ and state, $nmpreg_{ist}$ and x_{ist}, and $nmpreg_{ist}$ and year the identification assumption is that the propensity to marry following a nonmarital pregnancy does not vary with state, demographics, or year. This assumption can be relaxed by including the full set of interactions, and these specifications are reported in the main results table. However, the considerable number of interactions results in a loss of degrees of freedom, and, as will be seen in the next section, the main effect, while remaining similar in magnitude, loses statistical significance in the most flexible specification. For this reason, in the baseline regressions I rely on the stronger identifying assumption, but in the interest of both completeness and transparency I report the specifications requiring weaker assumptions.

Assessing the DID design

The differences-in-differences (DID) design is motivated by the idea that, after controlling for state and time effects, the precise timing and geography of the rollout of child support laws are uncorrelated with unobserved determinants of individual choices regarding marriage and fertility.

There are three main challenges for identification in this setting. The first is reverse causality: that changes in unobserved attitudes regarding marriage and fertility may be a

driving force behind child support statute adoption. For example, a declining attitude toward marriage may affect the desire to have stronger protections for children of unmarried couples, and therefore lead to stronger child support enforcement. I presented empirical evidence in Section 5.1 against this hypothesis. As a second check, I estimate a placebo regression that adds interactions between pregnancy and changes in a state's *future* child support law. This specification tests whether current marital choices predict future changes in child support, as they would if the reverse causality story were in effect.

The second concern for identification is that of omitted variables: if state-level time trends such as expanded work opportunities for women, or the absence of marriageable men due to incarceration, affects both marriage rates and trends in child support legislation, our estimates will be biased. To mitigate the concern that the estimated effect is driven by unobservables, I report results with and without a large set of time-varying controls. First, I control directly for societal attitudes at the region-year level, which proxy for public sentiment on issues including premarital sex, sex education, abortion, and the generosity of public assistance. Next, I add time-varying state controls, including percent black in the population, the male unemployment rate, the share of college graduates who are female, the female labor force participation rate, urban share of households, share of welfare recipiency, share of single moms on welfare, and the share of people below 100% of the federal poverty line. Third, I show that the estimates are robust to specifications allowing for differential time trends related to baseline state-level characteristics. Fourth, I show that the estimates are robust to the inclusion of state-year interacted fixed effects, along with interactions of all right-hand side variables with state dummies and time dummies. This last specification relies on variation across women in pregnancy status *within* state-year cells, and has the advantage of holding fixed all state-level time-varying determinants of marriage that may be correlated with the timing of child support enactments.

The third challenge for identification is that the timing of pregnancy is to some extent a choice, as is explicitly captured in the model of Section 3. Hence, Equation 5.4 faces

a selection concern. I defer this discussion until Section 5.4, after presenting estimates of the direct effect of child support enforcement on fertility. For now, it is worth emphasizing that in order to give the estimates of Equation 5.4 a causal interpretation, we must assume that child support laws do not cause differential selection into pregnancy based on underlying propensity to marry. In the model of Section 3, indeed there *is* differential selection based on propensity to marry, but the selection works against finding a negative coefficient β_2: couples who are least likely to marry decrease their fertility the most, implying that a selection effect would make child support appear to *increase* the propensity to marry.

Event study illustration

Figure 6 provides a graphical depiction of the identification strategy, represented as an event study. Since the child support index is a continuous variable, it is useful to recenter the data so that time 0 represents the biggest change in the child support index for each state, after removing state and year fixed effects. We then plot the outcome variables and controls relative to this "event." All figures discussed here include only the sample of states, 35 out of 48, for which we can construct a balanced time series of 3 years before and after the biggest change, so that changes in the series are not driven by changes in the composition of states included in the sample. Figures without this compositional adjustment look very similar and are presented in Appendix B. The child support series of all panels of Figure 6 show a sharp increase in the child support residual between time -1 and 0.[40]

To construct the shotgun marriage series, depicted in the left panel of Figure 6, I regress a marriage dummy (equal to one if the person marries in the current year or the next year) on a pregnancy dummy, along with state and year fixed effects, and individual controls (race dummies, age, age-squared, and a high school degree dummy), separately in each "recentered" year. The regression sample includes women in the 18-44 age cohort whose first pregnancy was outside of marriage. I then plot the coefficient estimate on the pregnancy dummy in

[40]Note that what appears to be a decline in the child support index approaching year 0 is a consequence of the regression mechanics, which forces the residuals to sum to zero across time periods.

each year, so that time 0 represents the year after the biggest jump in child support residual. We see from the figure that the increase in the child support law index, from year -1 to 0 is associated with a substantial decline in the probability of marriage following a nonmarital pregnancy.

The right panel of Figure 6 anticipates the analysis of abortion rates to come in the next section. To construct the abortion series, I first remove year and state fixed effects from the abortion rate. I then plot the mean of these residuals in each recentered year. Again, for the abortion rate series, time 0 represents 1 year after the biggest jump in the child support law index (stripped of state and year fixed effects). The figure shows a decline in the abortion rate following an increase in child support enforcement.[41]

We can contrast Figure 6 with event study plots using control variables. The research design requires no sharp changes in unobserved determinants of shotgun marriages around the "event." If, for example, sharp changes in female labor market opportunities took place at the same time as the child support rollout, we would be concerned about omitted variables. Figure 7 depicts the event study with several other variables on the left y-axis, after removing state and year fixed effects: the female labor force participation rate, the female college share, the fraction of residents on welfare, and the fraction of residents below the 100% federal poverty line. Note that there do not appear any sharp changes around the biggest change in child support laws. This is reassuring, and it lends support to the research design which requires common trends in unobserved determinants of shotgun marriages. Similar pictures for additional controls and for the GSS societal attitudes data are presented in Appendix Figures B.2 and B.3.

[41]Note that the child support series in the left panel is slightly different than its counterpart in the right panel. The reason is that the abortion data has gap years and hence the variation in the child support series used for identification is slightly different for the two outcome variables.

Marriage Results

Table 6 reports the results from the estimation of Equation 5.4. Column 1 reports estimates using state and year fixed effects. Column 2 includes region-year societal attitudes controls, including the principal factor from a factor analysis of the 9 abortion-related questions in the GSS, and four additional variables controlling for attitudes toward federal welfare, sex education, divorce, and pre-marital sex. Column 3 includes additional state-year controls. Column 4 includes baseline (1977) controls (including societal attitude controls) interacted with linear time trends. The results are remarkably stable across specifications. In the first four columns, the coefficient on the interaction term $nmpreg_{ist} \times CS_{st}$ is negative and significant, with estimates between $-.117$ and $-.120$. These coefficients indicate that a woman who has a nonmarital pregnancy in a state-year that has all child support laws in place compared to a nonmarital pregnancy in a state-year with none is about 12 percentage points less likely to have a shotgun marriage. These estimates are relative to baseline of 38 percent likelihood of becoming married in the same year or in the year following a nonmarital pregnancy, which can be seen in the coefficient in the second row.

Note that the regressions in columns 1-4 do not include interactions between $nmpreg_{ist}$ and state, x_{ist}, or year. This means that these regressions are estimated under the assumption that the propensity to marry following a nonmarital pregnancy does not vary with state, x_{ist}, or year, after conditioning on controls.

Column 5 reports the results includes interactions between $nmpreg_{ist}$ and state dummies, and $nmpreg_{ist}$ and x_{ist}, allowing flexibly for women in different states and demographics to have different propensities to marry following a nonmarital pregnancy. The point estimate is $-.154$ and is significant at the 1 percent level. Column 6 adds an interaction between $nmpreg_{ist}$ and a linear time trend, and the point estimate is $-.145$ and the estimate remains significant at the 10 percent level. The point estimate diminishes only slightly, mitigating concern for bias due to a violation of common trends. Note that the coefficient estimate on $nmpreg_{ist} * t$ is quite small in magnitude and statistically insignificant. This economically

small effect size lends support to the prior specifications that assume propensity to marry evolves in the same way for women who are pregnant and those who are not.

Column 7 adds interactions between $nmpreg_{ist}$ and all year dummies, flexibly allowing for propensity to marry following a pregnancy to vary with each year in the sample. The regression loses considerable power and the standard errors become large. The main effect is no longer significant, but the effect size remains similar in magnitude at $-.143$. An F-test of the null hypothesis that all interactions between $nmpreg_{ist}$ and year dummies are jointly equal to zero fails to reject the null ($F(20, 40) = 1.458$ and $p = .170$), mitigating concern that the main estimate is driven by time trends in the propensity to marry following a pregnancy.

Column 8 restricts the sample to pregnancy-year observations, meaning I restrict the sample to observations with $nmpreg = 1$. By restricting to these observations we are effectively allowing state and time trends in the propensity to marry following a nonmarital pregnancy. The coefficient on the interaction term – reported in the top row for ease of comparison with the estimates in other columns – remains negative but the standard errors are quite large and in this specification we fail to reject a null effect.

The results taken together suggest that moving from a regime with 0 of the child support laws in place to one with all of them in place leads to about a 12 percentage point reduction in the shotgun marriage rate, off a base of 38 percent. Over the sample period the policy index increased from an average of .13 (in 1977), to .79 (in 1992). Scaling the column 3 point estimate by the actual growth in the policy index over this period delivers an effect size of $.117 * (.79 - .13) = .077$ percentage points. This effect size is quite large relative to the ~ 10 point decline in the shotgun marriage rate over the 1977-1992 sample period.[42]

The magnitude of the estimated effect may appear large given the substantial amount of non-payment of child support. It is worth pausing to reflect on the plausibility of such

[42]Appendix Table B.7 examines heterogeneity by the type of policy adopted, re-estimating Equation 5.4 and replacing the policy index with a separate indicator for each policy, and replacing the policy index interaction term with interactions of each policy with the pregnancy indicator. The policies that are statistically and economically significant are the genetic testing and wage withholding provisions. Less important policies for the marriage decision are penalties for failure to pay (such as allowing liens on property).

large effects. First, it is important to note that the effect can be large due to a substantial number of couples on the margin of indifference between marriage and remaining single; in this case, a small reward for not marrying can have a substantial effect on aggregate marriage rates. The marriage decision following a nonmarital pregnancy was chosen for this analysis precisely because they are marginal marriages in which couples are likely to exhibit a high elasticity with respect to marital incentives. Second, the estimates reported here are consistent with what other authors have reported; for example, Rossin-Slater (2013) studies in-hospital voluntary paternity establishment and finds that for each additional paternity established there are 0.13 fewer parental marriages post-childbirth. In addition, if paternity establishment gives fathers greater custodial rights outside of marriage, as discussed in Appendix A, marriage decisions may be affected even if a large fraction of established fathers are subsequently delinquent on payments.

Marriage results: selection into fertility

Let us now return to the discussion of selection into fertility raised in Section 5.4. To restate the selection issue, it is plausible that child support *only* affects fertility and not marriage; under this interpretation, the effects on marriage observed in Section 5.4 are because individuals who are most affected in their fertility also happen to be those most likely to marry following a pregnancy.

First, it should be noted that the literature has found that young and low-educated men and women reduce their fertility the most in response to child support legislation, while older and more-educated individuals have little response (Aizer and McLanahan, 2006; Plotnick et al., 2007). But older and higher educated men and women have far higher rates of shotgun marriage than younger and less-educated, meaning selection works against finding a negative effect of child support on marriage. The theoretical model in this paper supports the prior literature: couples who are least likely to marry reduce their fertility the most, again suggesting that the empirical effects estimated on marriage in Section 5.4 are a lower

bound of the true effect.

To examine the sensitivity of the marriage estimates to selection bias, I follow Bharadwaj et al. (2013) and perform an exercise that offers pessimistic counterfactual scenarios, and report these results in Appendix Table 7. To construct the counterfactuals, I begin by assuming that the full rollout of child support laws reduced total pregnancies by 2 percent, a large effect. Next, I impute "excess births" that would have occurred absent the policy. Specifically, I impute an additional $\Delta CS_{st} \cdot .02$ births to single women in each state-year: these are the births that are not observed due to the policy change, under the selection story.[43] I then randomly assign higher and higher shotgun marriage rates to these imputed births – 60 percent, 70 percent, etc. – and re-estimate the main specification of the marriage regression (column 3 of Table 6). The exercise shows that even when imputing large selection effects, the main effect on marriage is only slightly diminished. Even under the counterfactual scenario in which these non-occurring pregnancies have 90 percent marriage rates, the estimates are statistically significant at the 10 percent level. The point estimates remain consistent with the interpretation that child support has reduced marriage following a nonmarital pregnancy.

6 Conclusion

This paper presents both a model and empirical evidence to understand the effect of child support policies on selection into marriage and fertility. The model shows that child support enforces commitment from men to future transfers in the event of a child, and crowds out the commitment role of marriage. Child support also reduces abortions because of the diminished cost of raising the child as a single mom. The empirical evidence uses variation in the state-year rollout of child support laws over the 1977-1992 period relative to the timing of nonmarital pregnancy; I show that the rollout led to a reduction in the probability

[43] Note that $\Delta CS_{st} \cdot .02$ is the appropriate fraction since it implies, once we sum over the years that the full rollout would have affected fertility rates by a fraction $\Sigma_{t=1972}^{1992} \Delta CS_{st} \cdot .02 = .02 \cdot (CS_{s,1992} - CS_{s,1992})$ in each state.

of having a shotgun marriage, and a decline in the abortion rate.

The main takeaway is that child support laws, which are intended to ensure that the non-custodial parent contributes to child rearing, have important consequences for selection into fertility and marriage. The magnitudes estimated in this paper suggest that the increase in child support enforcement can account for a substantial part of the long-run decline of shotgun marriages, and part of the recent fall in the abortion rate. Child support laws are thus central to understanding major trends in marriage and fertility in the second half of the 20th century, during which child support enforcement went from virtual non-existence to automatic and near-universal.

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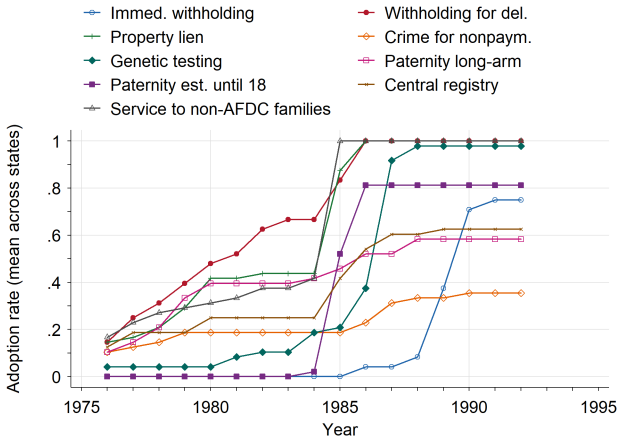
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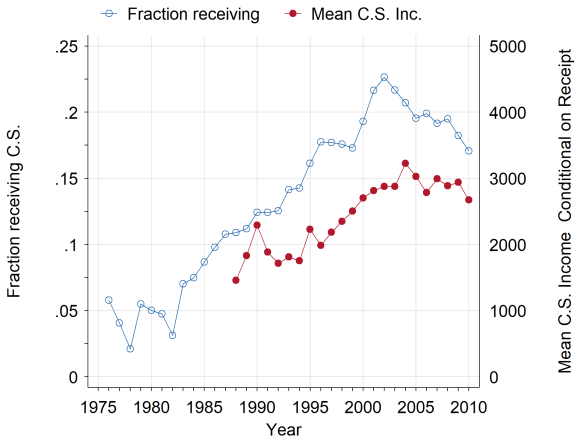
Figures and Tables

Figure 1: Child support statute adoption



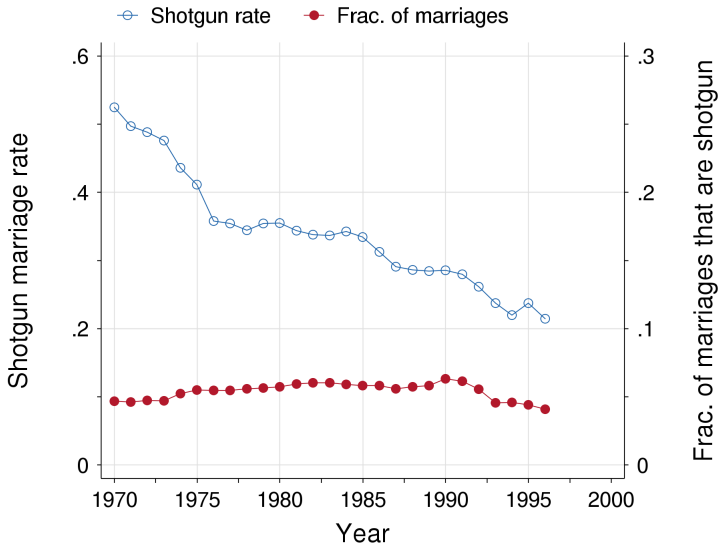
The figure above plots the mean rate of adoption across states for each of the 9 types of child support statutes used in the analysis.

Figure 2: Child support income receipt



The figure above plots two annual series: (1) the fraction of never-married mothers with absent father receiving any child support income, and (2) mean child support income received (in 2000 USD) for those receiving any child support. Source: March CPS.

Figure 3: Shotgun marriage rates by year



The figure uses data from restricted-SIPP and plots three-year moving averages of the following two series: (1) the fraction of first births that are resolved in a marriage 0-8 months before birth, plotted on the left y-axis, and (2) the fraction of first marriages that are shotgun marriages, plotted on the right y-axis.

Figure 4: Trends in abortions in the U.S.

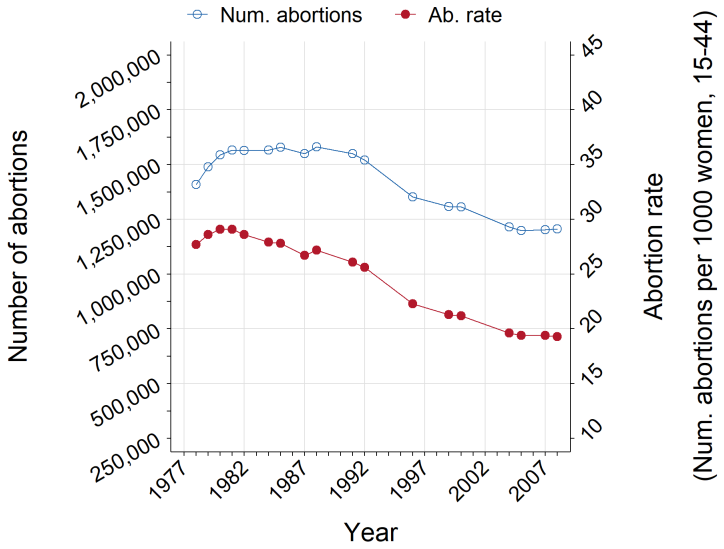
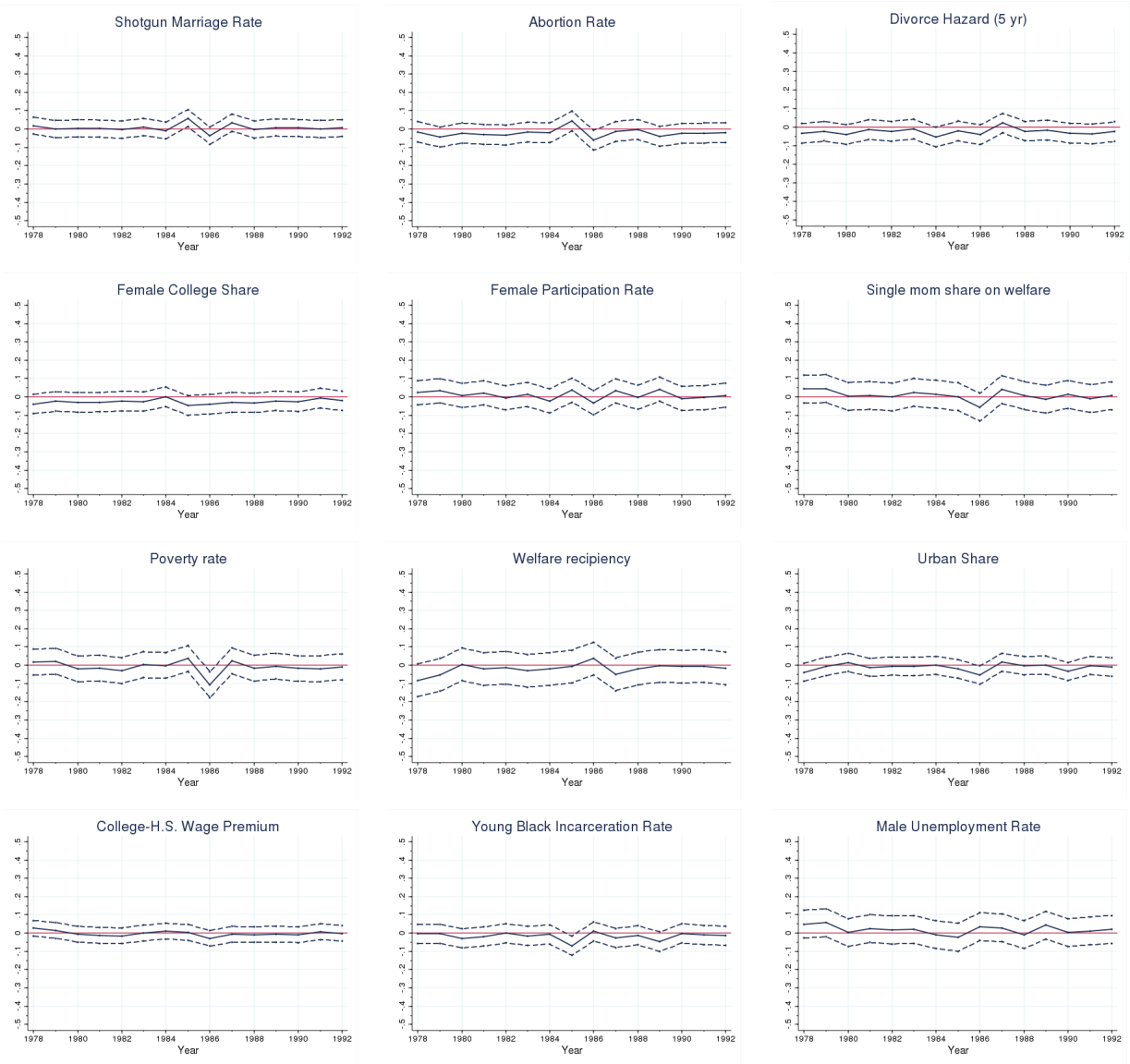
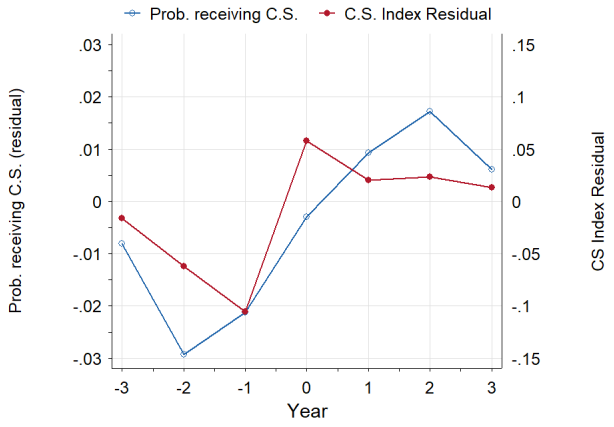


Figure 5: Child support legislation changes regressed on baseline state characteristics, 1977-1992

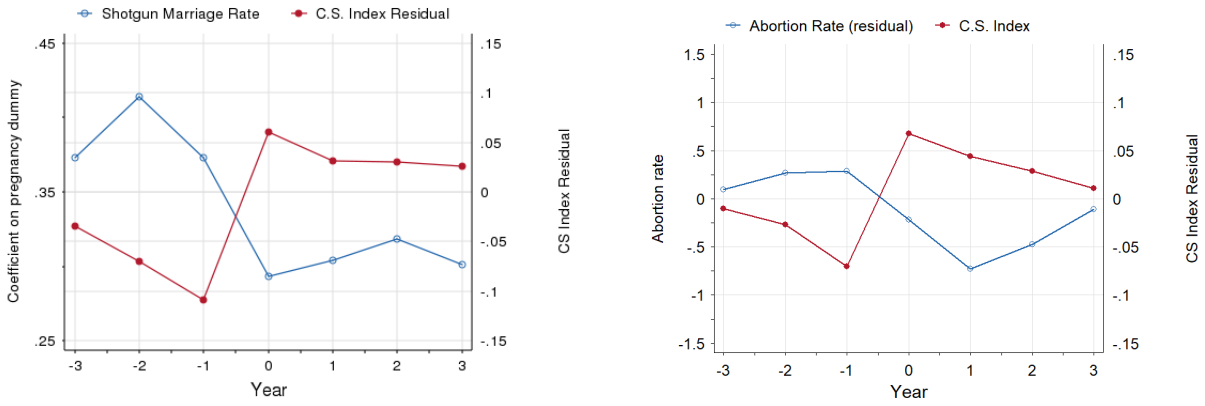


I regress changes in the child support index on state specific baseline characteristics interacted with time dummies, controlling for state and time fixed effects. The figures plot estimates $\hat{\psi}_t$ from the regression equation $\Delta CS_{st} = \alpha + \psi_t' w_{s,1977} + \eta_s + \xi_t + \epsilon_{st}$. The vector $w_{s,1977}$ are baseline (1977) characteristics from each state (except for the black incarceration rate, for which the baseline year is 1980, and the abortion rate, for which the baseline year is 1978), and $\Delta CS_{st} = CS_{st} - CS_{s,t-1}$. Each $w_{s,1977}$ is standardized to have mean zero and standard deviation 1.

Figure 6: Event study illustration of the DID design

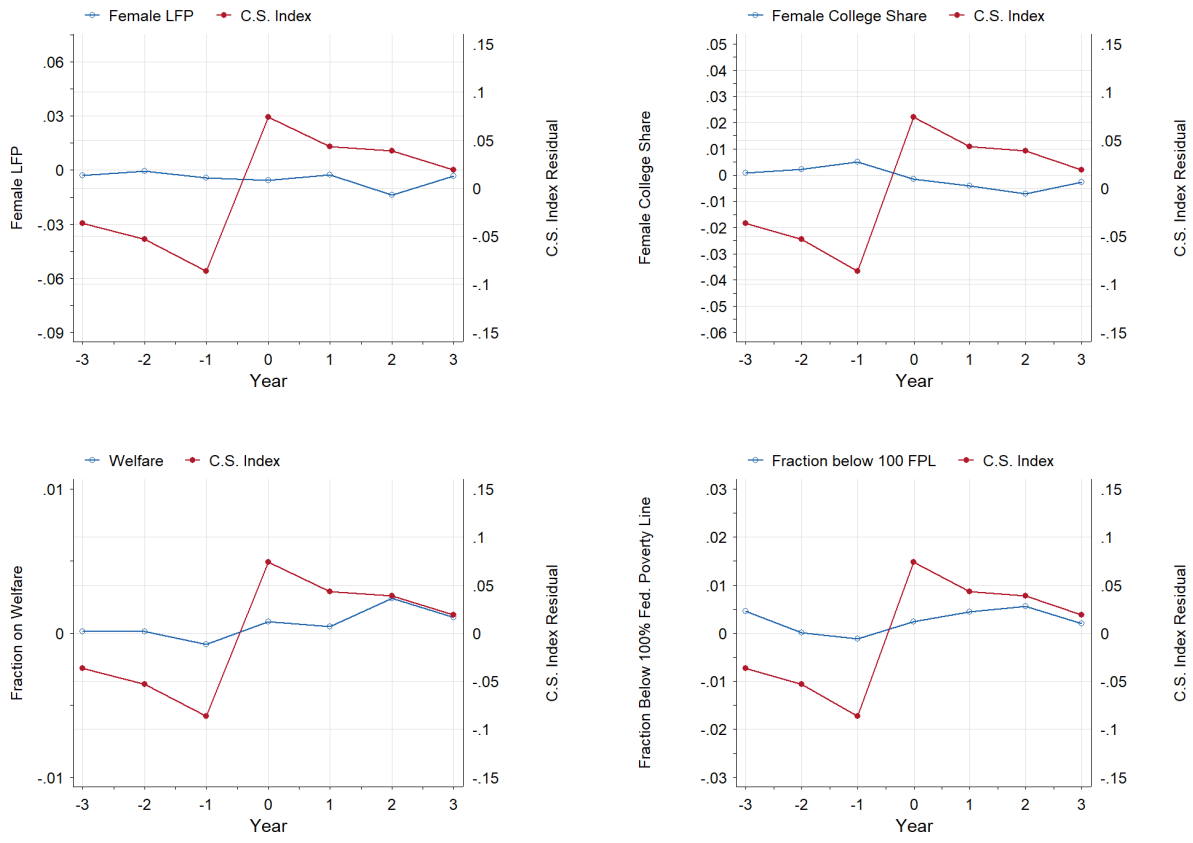


The above panel presents the first stage as an event study. For each state I recenter the data so that both cause and effect occur at time 0. For the child support policy index (right y-axis), time 0 represents the year with the strongest growth in the policy index in each state, after removing state and year fixed effects. For the child support income series (left y-axis), I regress the probability of receiving child support on controls, with state and survey year fixed effects. I then plot the mean residuals in the recentered year, where year 0 represents the year after the policy change.



The policy index series is constructed exactly as in the first stage panel above. For the marriage series and abortion series, time 0 represents the year after the biggest change in the policy index. The marriage series (left panel) is constructed as follows: in each recentered year, I regress the marriage dummy on a dummy for pregnancy and controls, for the sample of women who experience a nonmarital pregnancy; the figure plots the coefficients in each recentered year. The right panel plots mean residuals, in each recentered year, from a regression of the abortion rate on state and year fixed effects. The range of the right y-axis is 1 standard deviation of the child support index; the range of the left y-axis is one standard deviation of the abortion rate variable. The figures include 35 out of 48 states for which we can construct a balanced sample 3 years before and after the biggest jump in the legal index.

Figure 7: Event study illustration: other variables



This figure repeats the exercise of Figure 6 with alternative state-level variables depicted on the left y-axis. The range of the left y-axis is 2 standard deviations of the left y-axis variable.

Table 1: Descriptive statistics of the sample

	All yrs.	1977-1980	1981-1984	1985-1988	1989-1992
Person characteristics					
Shotgun marriage rate	0.36	0.38	0.39	0.36	0.33
	(0.48)	(0.48)	(0.49)	(0.48)	(0.47)
Fraction of marriages: shotgun	0.10	0.09	0.11	0.10	0.11
	(0.30)	(0.29)	(0.31)	(0.30)	(0.32)
Fraction of pregnancies: non-marital	0.31	0.26	0.29	0.32	0.35
	(0.46)	(0.44)	(0.45)	(0.47)	(0.48)
Fraction of births: non-marital	0.21	0.18	0.19	0.22	0.25
	(0.41)	(0.38)	(0.39)	(0.41)	(0.43)
Child Support					
Child Support Index	0.46	0.19	0.27	0.64	0.78
	(0.29)	(0.13)	(0.13)	(0.15)	(0.12)
C.S. Expend. (2000 USD)	223.62	179.07	179.65	218.94	287.87
	(111.82)	(95.53)	(92.22)	(102.67)	(113.00)
State-level controls					
Percent black	0.12	0.12	0.12	0.12	0.13
	(0.08)	(0.07)	(0.07)	(0.08)	(0.08)
Male Unemployment Rate	0.08	0.07	0.10	0.07	0.07
	(0.03)	(0.02)	(0.03)	(0.02)	(0.02)
Females w/ College Deg.	0.44	0.42	0.43	0.44	0.46
	(0.03)	(0.02)	(0.03)	(0.02)	(0.02)
Female LFP	0.64	0.59	0.63	0.66	0.69
	(0.06)	(0.04)	(0.04)	(0.04)	(0.04)
Urban	0.72	0.69	0.70	0.74	0.75
	(0.22)	(0.23)	(0.23)	(0.22)	(0.21)
Welfare	0.03	0.03	0.03	0.03	0.03
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Single mom welfare	0.10	0.10	0.10	0.10	0.09
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
Poverty	0.13	0.12	0.14	0.14	0.13
	(0.04)	(0.03)	(0.04)	(0.03)	(0.03)

This table reports the mean and standard deviations (in parentheses) of the marriage data from the restricted-SIPP, the child support legislative data, and the state-year controls. The shotgun marriage rate is defined as the fraction of nonmarital pregnancies (resulting in births) that are resolved in a marriage between 0 and 8 months prior to birth. Nonmarital pregnancy refers to the fraction of women with a nonmarital conception resulting in birth. Standard deviations are in parentheses.

Table 2: Descriptive statistics: Guttmacher, GSS, and State Legislature data

	All yrs.	1977-1980	1981-1984	1985-1988	1989-1992
Abortion statistics					
Num. abortions per 1000 women	27.68	28.65	28.56	27.24	25.86
	(10.57)	(9.82)	(10.55)	(11.08)	(10.67)
Num. abortion providers (per 100,000 women)	4.82	5.31	5.18	4.56	4.06
	(2.90)	(3.01)	(2.98)	(2.82)	(2.54)
Societal attitudes					
Abortion legal for any reason	0.43	0.40	0.42	0.41	0.47
	(0.11)	(0.12)	(0.12)	(0.10)	(0.11)
Abortion legal if not married	0.46	0.47	0.47	0.43	0.49
	(0.11)	(0.11)	(0.11)	(0.09)	(0.11)
Abortion legal if married	0.46	0.47	0.48	0.44	0.48
	(0.11)	(0.11)	(0.12)	(0.10)	(0.12)
Abortion legal if woman's health endangered	0.90	0.91	0.90	0.89	0.92
	(0.04)	(0.03)	(0.04)	(0.04)	(0.04)
Abortion legal if pregnant as a result of rape	0.82	0.81	0.82	0.80	0.85
	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
Abortion legal if strong chance of birth defect	0.81	0.83	0.81	0.79	0.82
	(0.07)	(0.05)	(0.05)	(0.07)	(0.09)
Abortion legal if low income	0.49	0.52	0.50	0.45	0.52
	(0.10)	(0.09)	(0.10)	(0.09)	(0.11)
Pre-marital sex OK	0.50	0.48	0.49	0.52	0.49
	(0.10)	(0.08)	(0.10)	(0.11)	(0.11)
Should divorce be easier to obtain	0.31	0.35	0.27	0.30	0.32
	(0.09)	(0.09)	(0.08)	(0.08)	(0.10)
Sex ed. in public schools	0.90	0.85	0.90	0.91	0.92
	(0.05)	(0.06)	(0.05)	(0.04)	(0.04)
U.S. spends too much on welfare	0.45	0.58	0.46	0.42	0.37
	(0.12)	(0.08)	(0.07)	(0.10)	(0.11)
State legislature comp. (House-Senate-Governor)					
D-D-D	0.38	0.53	0.38	0.36	0.27
	(0.49)	(0.50)	(0.49)	(0.48)	(0.44)
D-D-R	0.32	0.22	0.34	0.30	0.40
	(0.47)	(0.41)	(0.47)	(0.46)	(0.49)
D-R-D	0.14	0.08	0.10	0.18	0.21
	(0.35)	(0.28)	(0.30)	(0.38)	(0.40)
D-R-R	0.04	0.01	0.05	0.04	0.04
	(0.19)	(0.12)	(0.22)	(0.20)	(0.20)
R-R-R	0.04	0.05	0.05	0.03	0.02
	(0.19)	(0.22)	(0.21)	(0.18)	(0.15)
R-R-D	0.05	0.05	0.06	0.05	0.03
	(0.21)	(0.21)	(0.24)	(0.21)	(0.17)
R-D-R	0.02	0.02	0.02	0.03	0.01
	(0.14)	(0.15)	(0.15)	(0.16)	(0.09)
R-D-D	0.02	0.03	0.01	0.01	0.02
	(0.13)	(0.17)	(0.08)	(0.10)	(0.14)
Other	0.00	0.01	0.00	0.00	0.00
	(0.05)	(0.09)	(0.00)	(0.00)	(0.06)

This table reports the mean and standard deviations (in parentheses) of the Guttmacher data on abortions per 1000 women aged 15-44, the number of abortion providers per 100,000 women. Societal attitudes data are from the General Social Survey (GSS), for respondents 18-44. State Legislature composition data comes from Halvorsen and Jakobsen (2013). See Appendix B for GSS variable definitions.

Table 3: Child support laws' effect on child support receipt, (1977-1992)

	Any child support income					
	(1)	(2)	(3)	(4)	(5)	(6)
Child support law index	0.128*** (0.018)	0.140*** (0.014)	0.053*** (0.017)	0.052*** (0.017)	0.054*** (0.017)	0.053** (0.020)
Age at survey	0.029*** (0.003)	0.029*** (0.003)	0.022*** (0.002)	0.022*** (0.002)	0.022*** (0.003)	0.021*** (0.003)
Age at first birth	-0.005*** (0.001)	-0.005*** (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)
African-American	-0.050*** (0.013)	-0.068*** (0.011)	-0.065*** (0.011)	-0.065*** (0.011)	-0.065*** (0.011)	-0.065*** (0.011)
Other race/ethnicity	-0.039* (0.020)	-0.031 (0.020)	-0.033* (0.019)	-0.032* (0.019)	-0.023 (0.018)	-0.027 (0.020)
HS Degree	0.056*** (0.008)	0.049*** (0.007)	0.050*** (0.007)	0.049*** (0.007)	0.050*** (0.007)	0.049*** (0.008)
Some college	0.077*** (0.010)	0.078*** (0.011)	0.075*** (0.011)	0.075*** (0.011)	0.075*** (0.011)	0.077*** (0.011)
College degree	0.097*** (0.014)	0.097*** (0.013)	0.094*** (0.014)	0.094*** (0.014)	0.093*** (0.014)	0.091*** (0.014)
State f.e.		Yes	Yes	Yes	Yes	
Year f.e.			Yes	Yes	Yes	
State-year controls				Yes	Yes	
Baseline controls * t					Yes	
State-year f.e.						Yes
Observations	24,565	24,565	24,565	24,219	22,776	24,565
R^2	0.036	0.053	0.058	0.058	0.059	0.116
Mean of dep. var.	0.17	0.17	0.17	0.17	0.17	0.17

The sample is from the March CPS and is restricted to never married mothers between 17 and 60 with a child 18 years or less in the household whose father is reported absent from the household. The dependent variable is an indicator for whether the mother receives any child support income at time t. The right-hand side variable of interest is an index of child support laws in the mother's state in the year prior to the birth year of the youngest child, where the index is constructed using the Case-McLanahan data, 1977-1992. Controls included but not reported: age, age-squared, age at birth, race dummies education dummies, number of children dummies. Column (6) includes state-year interacted fixed effects. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table 4: The effect of child support laws on abortions (1978-1992)

	(1)	(2)	(3)	(4)	(5)	(6)
CS Index	-3.059**	-2.956**	-2.571**	-1.855**	-2.059*	-1.802
	(1.270)	(1.254)	(1.135)	(0.915)	(1.048)	(1.124)
Ab. attitudes control		Yes	Yes	Yes	Yes	Yes
State-Year Controls			Yes	Yes	Yes	Yes
Baseline controls * t				Yes	Yes	Yes
Ab. providers control					Yes	
Region * t						Yes
Number of observations	720	720	720	720	720	720
R^2	0.979	0.979	0.979	0.981	0.982	0.981
Mean of dep. var.	27.55	27.55	27.55	27.55	27.55	27.55

The dependent variable is number of abortions per 1000 women age 15-44, regressed on the child support law index. All specifications include state and year fixed effects. The abortion attitudes control is the principal factor from a factor analysis of the 9 abortion-related questions in the GSS, aggregated to the region-year level. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table 5: Bounding selection: counterfactuals using non-pregnancies due to child support

	Baseline	Percentile of abortion rate assigned to non-pregnancies				
		Median	60th	70th	80th	90th
CS Index	-2.571**	-2.129*	-2.062*	-2.043*	-2.001*	-1.859
	(1.135)	(1.138)	(1.139)	(1.139)	(1.140)	(1.141)
Ab. attitudes control	Yes	Yes	Yes	Yes	Yes	Yes
State-Year Controls	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	720	716	716	716	716	716
R^2	0.979	0.979	0.979	0.979	0.979	0.979
Mean of dep. var.	27.55	27.55	27.55	27.55	27.55	27.55

This table assigns counterfactual abortion rates to pregnancies that are not observed in the data due to a direct effect of child support on fertility. These pregnancies are assigned abortion probabilities at the percentile (indicated at each column) in the initial (1978) state-level distribution. The baseline specification comes from Table 4 column (3). Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table 6: Child support laws' effect on marriage (1977-1992)

	Marriage cohort of women							Preg-Yrs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
N.M. Pregnancy X C.S. Index	-0.120*** (0.040)	-0.120*** (0.040)	-0.117*** (0.040)	-0.118*** (0.041)	-0.154*** (0.031)	-0.145* (0.079)	-0.143 (0.122)	-0.079 (0.117)
N.M. Pregnancy	0.381*** (0.020)	0.381*** (0.020)	0.380*** (0.021)	0.381*** (0.021)	0.290*** (0.033)	0.292*** (0.038)	0.269*** (0.048)	
C.S. Index	0.007 (0.006)	0.010* (0.005)	0.012** (0.005)	0.011** (0.005)	0.012** (0.005)	0.012** (0.005)	0.012** (0.005)	
N.M. Pregnancy X t						-0.001 (0.005)		
Soc. Attitudes Controls		Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-Year Controls			Yes	Yes	Yes	Yes	Yes	Yes
State-year baseline * t				Yes	Yes	Yes	Yes	
N.M. Preg * st. dum.					Yes	Yes	Yes	
N.M. Preg * t						Yes		
N.M. Preg * yr dum.							Yes	
Number of individuals	54,500	54,500	54,000	53,000	53,000	53,000	53,000	5,000
R^2	0.070	0.071	0.070	0.071	0.077	0.077	0.077	0.159
Mean of dep. var.	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.41

Data is from 1992, 1993 and 1996 restricted-use SIPP, and the sample is limited to females in the 18-44 age cohort who were never-married as of 9 months before their first birth. The dependent variable is an indicator for the mother getting married for the first time in year t or $t+1$. All regressions include state and year fixed effects, a high school degree indicator, race dummies, and age dummies. Region-year societal attitudes controls come from the GSS and include: the principal factor of the 9 abortion-related questions, and four additional controls for attitudes toward welfare, sex education, divorce, and pre-marital sex. State-year controls include: percent black in the population, the male unemployment rate, the share of college graduates who are female, the female labor force participation rate, urban share of households, share of welfare recipiency, share of single moms on welfare, and the share of people below 100% of the federal poverty line. Column (4) includes baseline (1977) state and region controls interacted with linear time trends. Column (5) adds interactions between *nmpreg* and state, and *nmpreg* and x_{ist}; Column (6) adds an interaction between *nmpreg* and a linear time trend; Column (7) adds interactions between *nmpreg* and year dummies. Column (8) restricts the sample to nonmarital pregnancy-years only. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table 7: Placing bounds on selection: counterfactuals using non-pregnancies due to child support

	Baseline	Shotgun marriage rate assigned to imputed pregnancies			
	(1)	(2) 50 pct	(3) 60 pct	(4) 70 pct	(5) 90 pct
N.M. Pregnancy X C.S. Index	-0.117*** (0.040)	-0.112*** (0.039)	-0.112*** (0.040)	-0.111** (0.042)	-0.111** (0.041)
N.M. Pregnancy	0.380*** (0.021)	0.379*** (0.020)	0.380*** (0.020)	0.382*** (0.021)	0.387*** (0.021)
C.S. Index	0.009 (0.006)	0.009* (0.006)	0.010* (0.006)	0.010* (0.006)	0.010* (0.006)
Soc. Attitudes Controls	Yes	Yes	Yes	Yes	Yes
State-Year Controls	Yes	Yes	Yes	Yes	Yes
Number of individuals	54,000	54,000	54,000	54,000	54,000
R^2	0.070	0.071	0.071	0.072	0.072
Mean of dep. var.	0.05	0.05	0.05	0.05	0.05

This table assigns counterfactual marriage rates to pregnancies that are not observed in the data due to a direct effect of child support on fertility. I start by assuming the full child support rollout reduced nonmarital births by 2 percent. I then impute an additional $\Delta CS_{st} \cdot .02$ births to single women in each state-year. These pregnancies are randomly assigned marriage rates indicated at each column heading. The regression specification comes from Table 6 column (3), and all columns follow this specification. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

A Model Appendix

Model Solution

We begin with the period 2 decision.

First consider couples without a pregnancy ($k = 0$). Because children activate the only benefits of marriage, these couples will be indifferent between marriage and remaining single. In these cases of indifference, I assume that couples do not marry. Each partner will consume his or her income w_h, and enjoys utility $u(w_h)$.

Now, consider the period 2 decision in the event of a pregnancy ($k = 1$). We must consider three potential states: (1) the couple marries, (2) the couple does not marry and the mom raises the child on her own, (3) the couple does not marry and the mom aborts the pregnancy. Note that because children are the only benefit of marriage, the female will not abort the pregnancy if the couple marries; hence we need not consider this case. Note also that if the couple does not marry, the male partner will not make any transfers to the female, because there are no investments in children and he does not receive utility from the mother's private consumption.[1]

Women decide to abort the pregnancy if $v_F + \delta \leq u(w_h) - u(w_\ell)$. The left-hand side is individual-specific, so woman i will abort whenever $\gamma_i = v_F(i) + \delta(i) \leq \bar{\gamma} = u(w_h) - u(w_\ell)$. If she aborts the pregnancy, the male receives utility $u(w_h)$ and the female receives utility $u(w_h) - \delta(i)$. If she chooses not to abort the pregnancy and raises the child as a single mom, the male enjoys utility $u(w_h) + \alpha v_M$ and the female enjoys utility $u(w_\ell) + v_F$.

Now suppose the couple marries. Private consumption in marriage will depend on whether the female partner would abort the pregnancy absent the marriage, because it affects the partners' outside options. First, let us suppose she would abort the pregnancy.

[1] This modeling choice is for simplicity, and does not affect the qualitative implications of the model. See Willis (1999) for a model in which unmarried fathers make transfers to the mother.

The allocation problem in marriage is then

$$\max_{\{c_M \geq 0, c_F \geq 0\}} \quad u(c_F) + v_F + \mu[u(c_M) + v_M] \tag{A.1}$$

$$\text{subject to} \qquad c_F + c_M = w_h + w_\ell$$

$$\text{and} \qquad u(c_M) + v_M \geq u(w_h) \tag{A.2}$$

$$\text{and} \qquad u(c_F) + v_F \geq u(w_h) - \delta \tag{A.3}$$

where μ is the Pareto weight representing the outcome of a generalized bargaining problem. The second line is the budget constraint, while the third and fourth lines represent participation constraints for the male and female, respectively. The participation constraints require the male and female to receive at least as much utility inside marriage as outside it. Denote $\{c_M^*, c_F^*\}$ as the solution to Equation A.1. The value of marriage to men is $u(c_M^*) + v_M$; the value of marriage to women is $u(c_F^*) + v_F$. Note that for Equation A.1 to have a solution it must be the case that $v_M > 0$.

Now consider the consumption problem in the case where the female partner would *not* abort the pregnancy absent the marriage. The allocation problem is identical to Equation A.1 with the exception of the participation constraints. The male's participation constraint is now $u(c_M) + v_M \geq u(w_h) + \alpha v_M$. Note that he has more bargaining power within marriage because if the couple chooses not to marry he can still enjoy some utility from the child. The female's participation constraint is now $u(c_F) + v_F \geq u(w_\ell) + v_F$, or equivalently $c_F \geq w_\ell$. Denote $\{\tilde{c}_M, \tilde{c}_F\}$ as the solution to the consumption problem in this case, and the values are $u(\tilde{c}_F) + v_F$, and $u(\tilde{c}_M) + v_M$.

Now we turn to the first period. For men, the only potential cost of pregnancy arises if he has $v_M < 0$, and this becomes a cost only if he is matched with a partner who will not abort. These men will propose premarital sex as long as $v_M \geq -\frac{b}{\beta\lambda\alpha}$. All other men unambiguously benefit from premarital sex and thus will propose it to the female partner. Will she accept?

The female partner knows fully well in the first period whether she is matched with a man

who will marry in the event of a pregnancy. Consider first a woman who would abort absent a marriage. If she is matched with a man who will marry, she will accept premarital sex iff $b + \beta\lambda(u(c_F^*) + v_F - u(w_h)) \geq 0$; her decision depends on the value of sex, her consumption in marriage, her utility from children, and the wage losses from pregnancy. If she is matched with a man who will not marry, she will accept premarital sex iff $\delta \leq \frac{b}{\beta\lambda}$, i.e. if the psychic or monetary cost of the abortion is not too high.

Consider now a female who would raise a child as a single mom rather than abort. If she is matched with a man who can credibly commit, she will accept premarital sex iff $b+\beta\lambda(u(\tilde{c}_F)+v_F-u(w_h)) \geq 0$. This inequality generates a cutoff in v_F, which we denote $\underline{v}_F$, above which she will accept, and below which she will reject the offer. If she is matched with a man who cannot commit, she will accept premarital sex iff: $b+\lambda(u(w_\ell)+v_F-u(w_h)) \geq 0$. This inequality leads to a second acceptance threshold $\bar{v}_F = \frac{-b}{\beta\lambda} + u(w_h) - u(w_\ell)$. Weakly fewer women will accept premarital sex with a man who cannot commit, because she receives less consumption in the event of a child; hence, $\bar{v}_F \geq \underline{v}_F$. This leads to three segments of women: those who accept sex without commitment ($v_F \geq \bar{v}_F$), those who accept sex only with commitment ($\bar{v}_F > v_F \geq \underline{v}_F$), and those who never accept ($v_F < \underline{v}_F$).

Note that men who cannot commit to marriage ($v_M < 0$) are willing to transfer some resources to their female partners in the first period to persuade them to have premarital sex. But they cannot borrow, and their commitment to transfer resources to their partner in the next period is not credible. These men will walk away in period 2 in the event of a pregnancy.

Proof. Proposition 1.

In period 2, couples with a pregnancy decide between three states: (1) the couple marries; (2) the couple doesn't marry and the mom raises the child alone; (3) the couple doesn't marry and the female aborts the pregnancy. Note that couples who arrive at period 2 with a pregnancy will not be affected in their decision to marry because the total surplus of marriage is unchanged by the transfer.

Consider the case where the couple does not marry and the female partner must decide between between (2) and (3). The value of raising the child as a single mom is strictly greater, and the option to abort the child is less attractive. This can be seen by examining the trade off between these two states under child support: $u(w_h) - \delta \geq u(w_\ell + \tau) + v_F$. Strictly fewer women will abort a pregnancy; this proves Proposition 1.1.

Consider the value of marriage v. the female's outside option. The female's participation constraint in marriage is now $u(c_F) + v_F \geq u(w_\ell + \tau) + v_F$. Women whose participation constraint binds will thus have weakly higher consumption as a result of the child support policy. Her male counterpart's participation constraint is now $u(c_M) + v_M \geq u(w_h - \tau)$, and he will enjoy weakly less consumption within marriage. Child support does not affect the participation constraint for women who will abort the pregnancy absent marriage. This proves Proposition 1.2.

Consider now the premarital sex decision. Under child support, strictly more women are willing to have sex without commitment: the threshold $\bar{v}_F$ shifts down to $\bar{v}_F = \frac{-b}{\beta\lambda} + u(w_h) - u(w_\ell + \tau)$. The threshold making her indifferent between sex with commitment and no sex, $\underline{v}_F$, shifts down as well so now fewer women reject sex; this threshold shifts down weakly *less* than $\bar{v}_F$.[2] Women who will abort a pregnancy (even under child support) are unaffected in their premarital sex decision.

Now consider men. Without child support, every man with $v_M \geq -\frac{b}{\beta\lambda\alpha}$ proposes premarital sex to his partner. With child support, there is an additional cost. Men who will marry their partner will continue to propose premarital sex. Men who will not marry and who are matched with a woman who will abort the pregnancy will also continue to propose premarital sex. Consider men who will not marry and who are matched with a woman who will not abort. These men will continue to offer premarital sex as long as $v_M \geq -\frac{b}{\beta\lambda\alpha} + \frac{u(w_h) - u(w_h - \tau)}{\alpha}$; hence, the threshold is strictly higher and fewer men offer premarital sex.

[2]The threshold $\underline{v}_F$ shifts down less than $\bar{v}_F$ for two reasons: (1) concavity of the utility function means that τ is worth less to a woman who would otherwise consume w_ℓ, compared to one who would otherwise consume c_F^* ; (2) consumption in marriage can increase for women by *at most* τ, which only occurs if her participation constraint binds both with and without the child support reform.

Under child support we can develop an expression for the total number of pregnancies. There will be fewer women requiring a marriage commitment to have premarital sex, and fewer women rejecting sex regardless of commitment; these forces increase the number of pregnancies. But there are a greater number of men who will not offer premarital sex. This proves Proposition 1.3.

By assumption, some men with distaste for children ($v_M < 0$) are still willing to have pre-marital sex under child support. These men are described by the set $v_M \in [-\frac{b}{\beta\lambda\alpha} + \frac{u(w_h)-u(w_h-\tau)}{\alpha}, 0]$. These men who are matched with women with $v_M \geq \bar{v}_F$ will now have pre-marital sex without a marriage commitment. By allowing these men to commit *ex ante* to future transfers, which they could not do on their own, child support leads to more pre-marital sex without marriage among these couples. This proves Proposition 1.4. □

Some extensions

Increasing custodial rights to fathers

Suppose that in addition to requiring a transfer τ from the father, child support grants the father additional custodial rights. This assumption has been incorporated into marriage models (Rossin-Slater, 2013; Edlund, 2013) and reflects that prior to child support, if the father is not married to the mother he has no presumption of paternity by the state, and hence no rights or obligations to the child.

Suppose we increase α to $\tilde{\alpha} \in (0, 1)$. Consider period 2. For couples in which the mom would abort the pregnancy absent marriage, the surplus of marriage is: $S(v_M) = v_M + v_F + \delta$, and does not depend on α. For couples in which the mom prefers being a single mom, the surplus of marriage is $S(v_M) = (1 - \tilde{\alpha})v_M$ and couples will continue to marry if $v_M > 0$.

Consider the further extension of adding an additional i.i.d. match quality shock θ at the beginning of period 2. The marriage surplus will become $S(v_M) = (1 - \tilde{\alpha})v_M + \theta$, and couples will marry if $\theta > -(1 - \tilde{\alpha})v_M$. Hence, as $\tilde{\alpha}$ increases, marriages are less likely. The intuition is simple: by increasing the father's custodial rights outside of marriage, marriage

becomes less attractive. In period 1, there will be fewer men with $v_M < 0$ willing to have premarital sex because it commits them to spending time with a child they derive negative utility from. Hence there will be fewer unwanted children.

Increasing maternity benefits

Suppose the government imposes a flat tax rate τ_m on all individuals' wages, and rebates the tax revenue to women who have children (regardless of their marital status). Now women with children receive $w_\ell + \frac{w_h \tau_m}{n_c}$, where n_c is the equilibrium fraction of women with children. Note that this policy too will have a negative effect on the abortion rate; being a single mom becomes relatively more attractive. Her bargaining power within marriage is also improved by maternity benefits.

Now consider period 1. As with child support, under a maternity benefit women are more willing to have sex without commitment. Unlike the child support case, however, the man's decision to have premarital sex is unaffected by the maternity benefits policy. The reason is that he must pay taxes for maternity benefits regardless of whether he has a child.

Hence there are two key differences between a maternity benefits policy and a child support policy. The first is that total births unambiguously increase under maternity benefits, while under child support the effect is ambiguous. Second, under child support it is precisely those men who get disutility from children who have less of them. Hence child support has the effect of reducing unintended or unwanted births from men.

I stress that in a model with endogenous labor supply or investment in children, maternity benefits will generate more nuanced predictions. For example, if maternity benefits only accrue to women who do not work, or if maternal labor supply diminishes the available time for investments in children. I leave these important considerations aside, but they may present additional contrasts with a child support policy.

B Appendix: Additional Tables and Figures

Table B.1: GSS Data: Variable Definitions

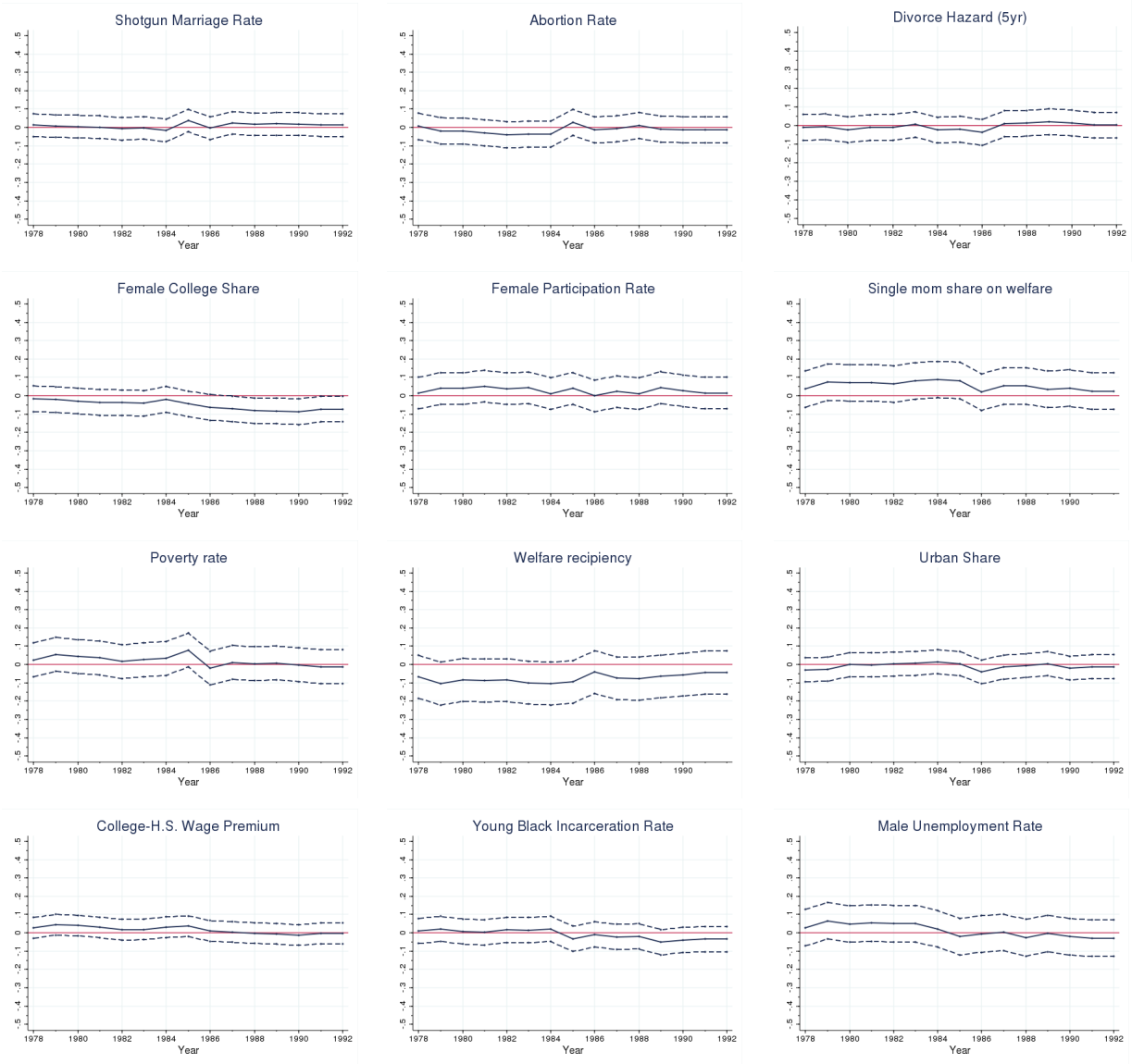
Survey question	**Notes**
Abortion Please tell me whether or not you think it should be possible for a pregnant woman to obtain a legal abortion if. . . (1) The woman wants it for any reason? (2) If she is married and does not want any more children? (3) If she is not married and does not want to marry the man? (4) If she is married and does not want any more children? (5) If the woman"s own health is seriously endangered by the pregnancy? (6) If she became pregnant as a result of rape? (7) If there is a strong chance of serious defect in the baby? (9) If the family has a very low income and cannot afford any more children?	All abortion survey responses: Yes=1, No, don't know, no answer=0. Years: 1977, 1978, 1980, 1982-1985, 1987-1991
Sex, marriage, welfare	
(1) Are we spending too much, too little, or about the right amount on welfare?	Too much=1; Too little, about right, don't know, no answer=0 Years: 1977, 1978, 1980, 1982-1991
(2) Would you be for or against sex education in the public schools?	Favor=1; Oppose, Depends, don't know, no answer=0 Years: 1977, 1982, 1983, 1985, 1986, 1988-1991
(3) Should divorce in this country be easier or more difficult to obtain than it is now?	Easier=1; More difficult, stay same, don't know, no answer=0 Years: 1977, 1978, 1982, 1983, 1985, 1986, 1988-1991
(4) There's been a lot of discussion about the way morals and attitudes about sex are changing in this country. If a man and woman have sex relations before marriage, do you think it is always wrong, almost always wrong, wrong only sometimes, or not wrong at all?	Not wrong at all=1; Almost always wrong, sometimes wrong, not wrong at all, don't know, no answer=0 Years: 1977, 1978, 1982, 1983, 1985, 1986, 1988-1991

Table B.2: State-year characteristics and the child support rollout

	(1)	(2)	(3)
Percent black		0.020	0.028
		(0.021)	(0.032)
Male		0.010**	-0.074*
unemployment rate		(0.005)	(0.039)
Female college		-0.004	-0.053*
share		(0.004)	(0.028)
Female		0.001	-0.049
participation rate		(0.010)	(0.038)
Welfare recip.		0.019**	0.020
		(0.007)	(0.042)
Welfare recip.		-0.002	-0.008
(single mom)		(0.006)	(0.037)
Poverty rate		0.006	-0.016
		(0.010)	(0.035)
College premium		-0.003	-0.021
		(0.004)	(0.025)
State f.e.	Yes	Yes	
Year f.e.	Yes	Yes	
Observations	768	768	48
R^2	0.917	0.920	0.176

This table presents regressions of the child support policy index on state and year fixed effects, along with time-varying controls that are likely to covary with marital attitudes or marital opportunities. In columns 1 and 2, the dependent variable is CS_{st}, and in column 3 the dependent variable is $CS_{s,1992} - CS_{s,1977}$. Controls are described in the text and are standardized to have mean zero, standard deviation 1 across state-year cells. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Figure B.1: Child support legislation levels regressed on baseline state characteristics, 1977-1992



This figure re-estimates Figure 5 with the dependent variable in levels rather than changes. Concretely, the figures plot the estimates $\hat{\psi}_t$ from the regression equation $CS_{st} = \alpha + \psi_t' w_{s,1977} + \eta_s + \xi_t + \epsilon_{st}$. The vector $w_{s,1977}$ are baseline (1977) characteristics from each state (except for the black incarceration rate, for which the baseline year is 1980, and the abortion rate, for which the baseline year is 1978). Each $w_{s,1977}$ is standardized to have mean zero and standard deviation 1.

Table B.3: Determinants of child support policy innovations: F-tests

	Shotgun rate	Ab. rate	Divorce (10yr)	Prison	Pct. black	Male unemp.	Fem. coll. share
F(20, 400)	1.420	1.378	0.879	1.072	0.533	0.655	0.463
p-value	0.134	0.155	0.588	0.381	0.922	0.829	0.958

	Fem. LFP	Urban	Welf.	Welf. mom	Pov. rate	Coll.-H.S. prem
F(20, 400)	0.825	1.095	0.715	0.794	1.574	0.678
p-value	0.650	0.359	0.769	0.685	0.078	0.806

	All coefficients
F(20, 400)	0.794
p-value	0.966

This table reports the F-tests of Equation 5.1 with the null hypothesis that the coefficients of variable $w_s^{(k)}$ are jointly equal to 0, i.e. $\psi_{1978}^{w_s^{(k)}} = \psi_{1979}^{w_s^{(k)}} = \cdots = \psi_{1992}^{w_s^{(k)}} = 0$.

Table B.4: Do societal attitudes toward abortion predict abortion availability?

	(1)	(2)	(3)	(4)
Abortion	7.213***	8.439***	0.193	0.404
attitudes factor	(1.485)	(1.734)	(0.360)	(0.354)
State			Yes	Yes
Year		Yes		Yes
Number of observations	765	765	765	765
R^2	0.286	0.364	0.935	0.969
Mean of dep. var.	4.74	4.74	4.74	4.74

The dependent variable is the number of abortion providers per 100,000 women aged 15-44. I regress this variable on the societal attitudes principal factor, which is constructed from factor analysis of the 9 abortion-related questions in GSS, which are described in Table B.1. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table B.5: Child support laws' effect on public assistance and welfare income by single moms, (1977-1992)

	Public assistance and SSI income			
	(1)	(2)	(3)	(4)
Child support law index	-245.547	-255.790	-502.994**	-351.596
	(204.017)	(188.895)	(189.128)	(218.679)
Age at survey	-5.149	-10.643	-28.490	87.976***
	(28.322)	(24.727)	(21.962)	(14.605)
Age at first birth	66.426***	65.814***	74.070***	120.035***
	(9.634)	(9.856)	(10.275)	(7.784)
African-American	271.972*	275.372*	278.549*	48.938
	(147.344)	(140.264)	(139.020)	(34.829)
Other race/ethnicity	69.160	77.067	44.270	-2562.188***
	(192.082)	(194.108)	(201.417)	(56.948)
HS Degree	-825.786***	-815.347***	-829.874***	-1005.703***
	(102.932)	(98.003)	(97.471)	(38.966)
Some college	-1200.658***	-1181.924***	-1180.844***	-1630.578***
	(104.454)	(105.545)	(107.150)	(34.367)
College degree	-1775.945***	-1761.924***	-1751.755***	-1521.300***
	(169.111)	(167.757)	(173.555)	(35.096)
State f.e.	Yes	Yes	Yes	
Year f.e.	Yes	Yes	Yes	
State-year controls		Yes	Yes	
Baseline controls * t			Yes	
State-year f.e.				Yes
Observations	24,565	24,219	22,776	24,565
R^2	0.300	0.310	0.320	0.389
Mean hhinc. (2000 USD)	25954.12	25954.12	25954.12	25954.12

The sample is from the March CPS and is restricted to never married mothers between 17 and 60 with a child 18 years or less in the household whose father is reported absent from the household. The dependent variable is the sum of pre-tax SSI income and public assistance/welfare income from the state (in 2000 USD) received by the mother at time t. The right-hand side variable of interest is an index of child support laws in the mother's state in the year prior to the birth year of the youngest child, where the index is constructed using the Case-McLanahan data, 1977-1992. Controls included but not reported: age-squared, number of children dummies. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table B.6: Child support laws' effect on marriage (1977-1992)

	Marriage cohort of women							Preg-Yrs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
N.M. Pregnancy X	-0.076***	-0.077***	-0.074**	-0.075**	-0.115***	-0.114*	-0.092	-0.009
C.S. Index	(0.028)	(0.028)	(0.028)	(0.028)	(0.028)	(0.065)	(0.116)	(0.109)
N.M. Pregnancy	0.369***	0.369***	0.369***	0.369***	0.323***	0.324***	0.306***	
	(0.016)	(0.016)	(0.016)	(0.016)	(0.026)	(0.029)	(0.042)	
C.S. Index	0.002	0.005	0.008	0.010**	0.011**	0.011**	0.011***	
	(0.006)	(0.005)	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)	
Soc. Attitudes Controls		Yes	Yes	Yes	Yes			Yes
State-Year Controls			Yes	Yes	Yes	Yes	Yes	Yes
State-year baseline * t				Yes	Yes	Yes	Yes	
N.M. Preg * st. dum.					Yes	Yes	Yes	
N.M. Preg * t						Yes		
N.M. Preg * yr dum.							Yes	
Number of individuals	54,500	54,500	54,000	53,000	53,000	53,000	53,000	5,000
R^2	0.069	0.069	0.069	0.069	0.074	0.074	0.074	0.141
Mean of dep. var.	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.44

This table replicates Table 6 without reweighting individuals to adjust for multiple appearances in the sample, as described in Section 5.4. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table B.7: Heterogeneity by policy: Child support laws' effect on marriage (1977-1992)

	Marriage cohort of women			
	(1)	(2)	(3)	(4)
Pat. Establishment				
Genetic testing	-0.056**	-0.056**	-0.056**	-0.056**
	(0.023)	(0.023)	(0.023)	(0.024)
Paternity	0.011	0.011	0.012	0.010
long-arm	(0.025)	(0.025)	(0.025)	(0.025)
Paternity est.	0.033	0.033	0.034	0.034
until 18	(0.022)	(0.022)	(0.022)	(0.023)
Withholding and Facilitating Payments				
Immed.	-0.032	-0.032	-0.031	-0.031
withholding	(0.029)	(0.029)	(0.029)	(0.030)
Withholding	-0.065*	-0.065*	-0.066*	-0.065*
for del.	(0.034)	(0.034)	(0.035)	(0.035)
Service to	-0.027	-0.027	-0.030	-0.030
non-AFDC families	(0.032)	(0.032)	(0.032)	(0.032)
Central registry	0.015	0.015	0.016	0.014
	(0.025)	(0.025)	(0.025)	(0.025)
Penalties for non-payment				
Property lien	0.011	0.011	0.012	0.013
	(0.037)	(0.037)	(0.037)	(0.037)
Crime for	-0.002	-0.002	0.001	0.001
nonpaym.	(0.032)	(0.032)	(0.033)	(0.033)
Soc. Attitudes Controls		Yes	Yes	Yes
State-Year Controls			Yes	Yes
State-year baseline * t				Yes
Number of individuals	54,500	54,500	54,000	53,000
R^2	0.071	0.071	0.071	0.071
Mean of dep. var.	0.06	0.06	0.06	0.06

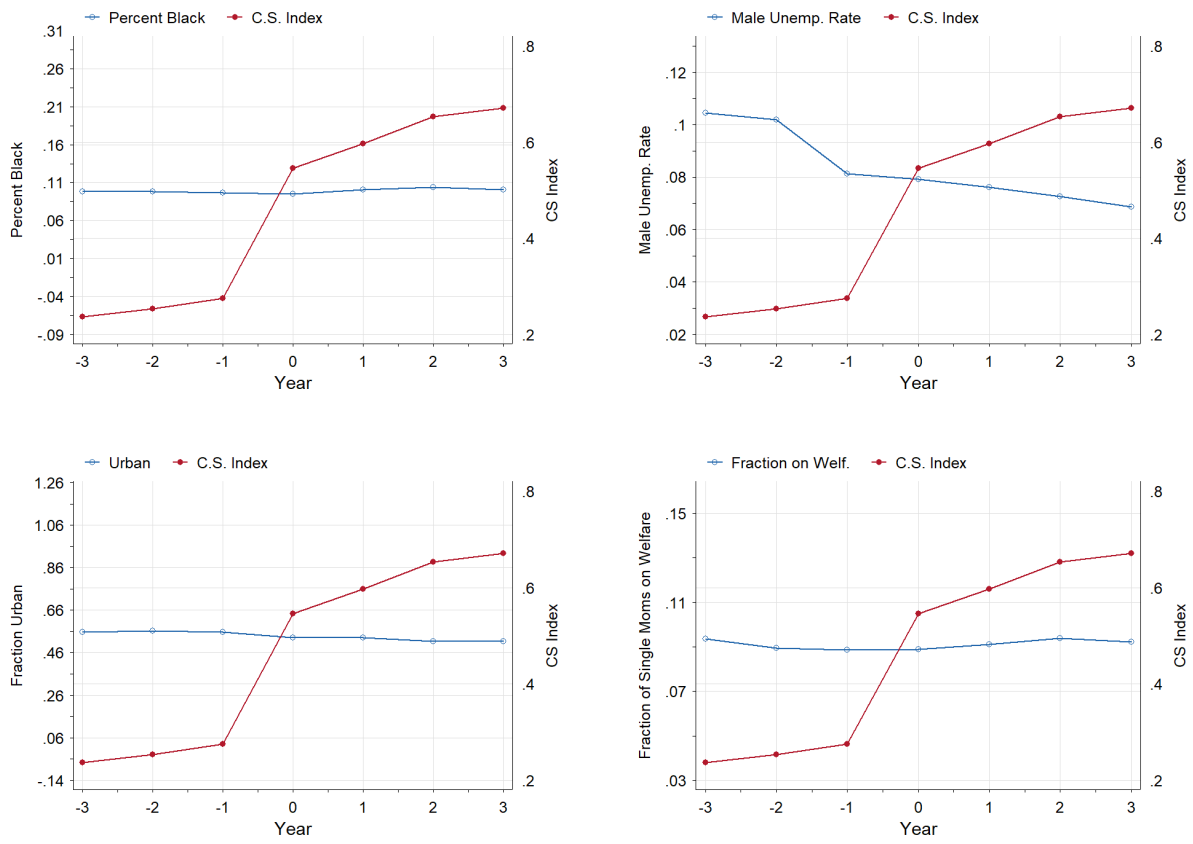
This table replicates Table 6 Columns 1-4, except includes each of the 9 law types separately and interacted with the pregnancy indicator. Standard errors are clustered at the state level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B.8: The effect of child support laws on abortions (1978-1992): non-imputed data

	(1)	(2)	(3)	(4)	(5)	(6)
CS Index	-3.360**	-3.306**	-3.074**	-2.267**	-2.322**	-2.081
	(1.262)	(1.265)	(1.230)	(1.082)	(1.144)	(1.347)
Ab. attitudes control		Yes	Yes	Yes	Yes	Yes
State-Year Controls			Yes	Yes	Yes	Yes
Baseline controls * t				Yes	Yes	Yes
Ab. providers control					Yes	
Region * t						Yes
Number of observations	384	384	384	384	384	384
R^2	0.975	0.975	0.976	0.977	0.978	0.977
Mean of dep. var.	27.55	27.55	27.55	27.55	27.55	27.55

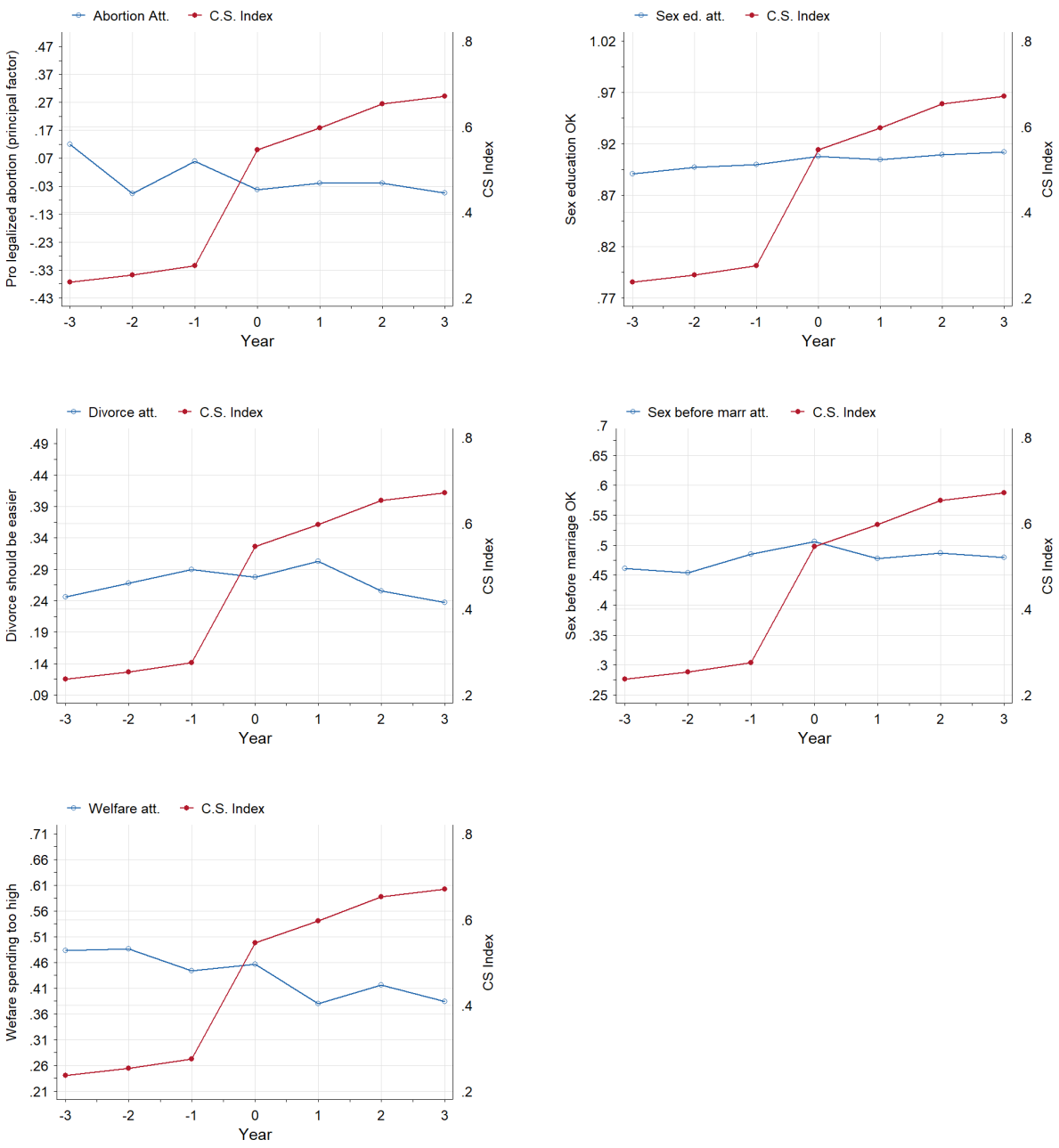
This table replicates Table 4 using only the non-imputed abortion rate data and the non-imputed GSS abortion attitudes data. Standard errors clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

Figure B.2: Event study illustration: other variables



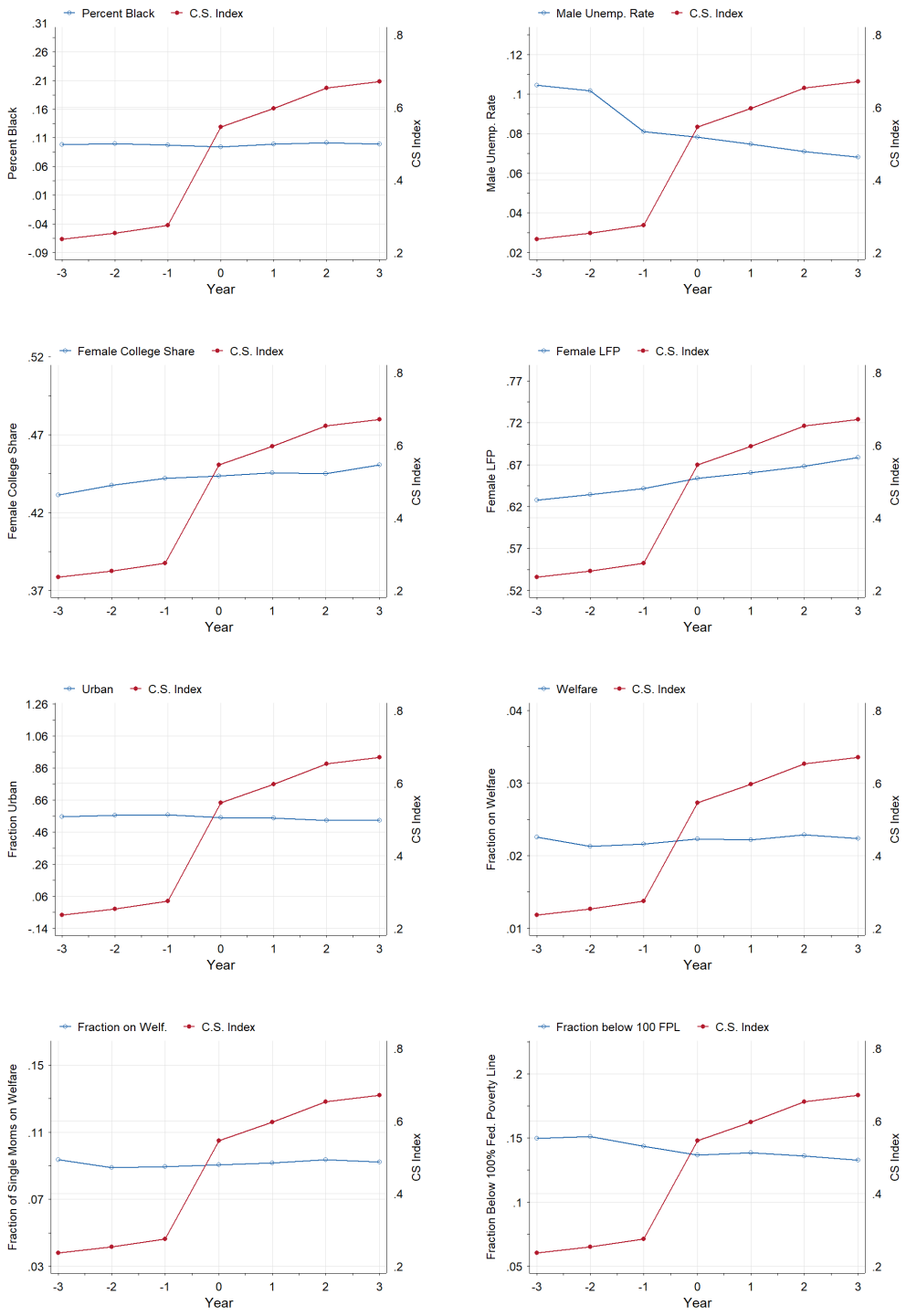
This figure repeats the exercise of the left panel of Figure 6 with alternative state-level variables depicted on the left y-axis. The range of the left y-axis is 2 standard deviations of the left y-axis variable.

Figure B.3: Event study illustration: GSS attitudes



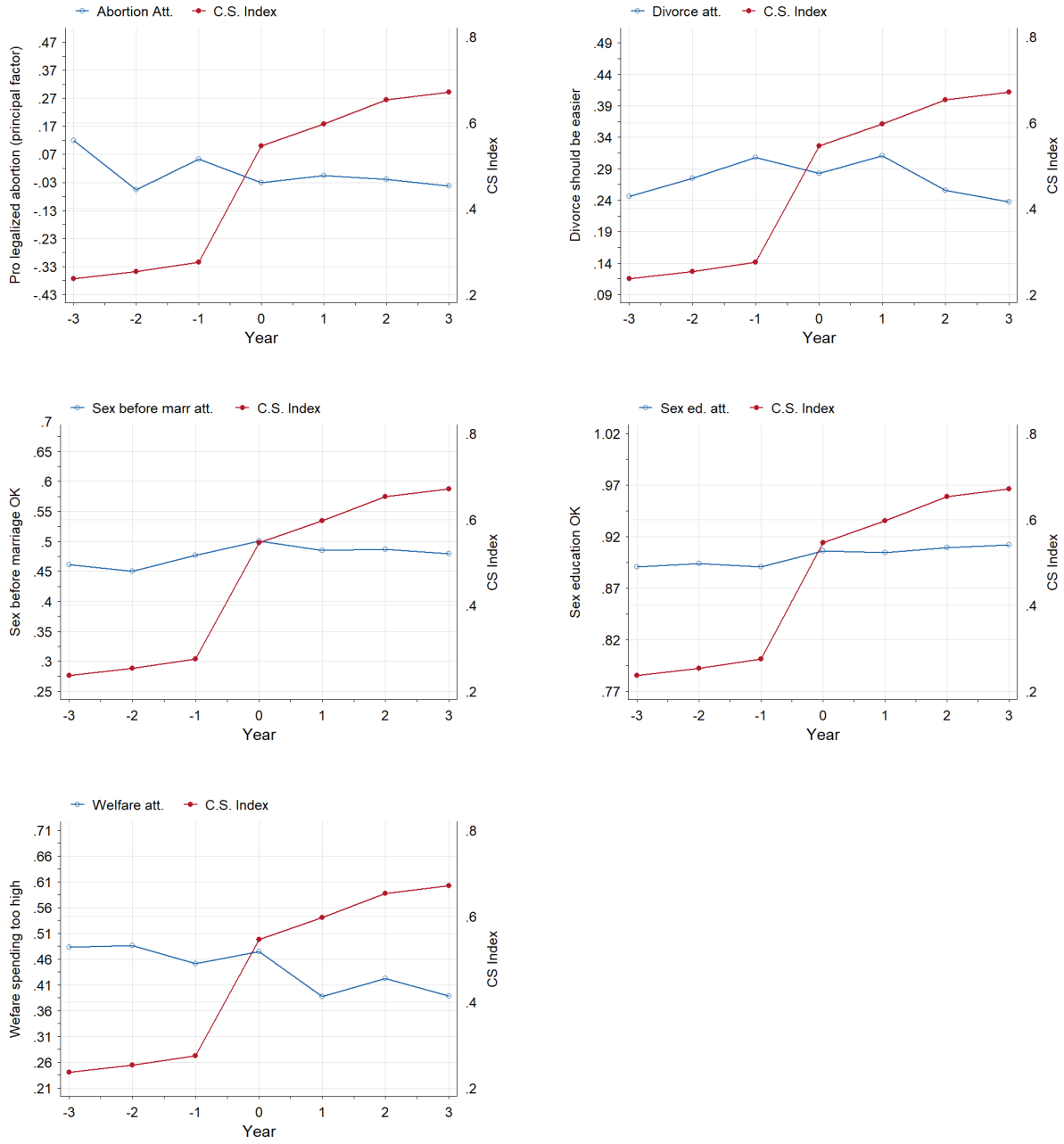
This figure repeats the exercise of the left panel of Figure 6 with GSS region-level variables depicted on the left y-axis. The range of the left y-axis is 2 standard deviations of the left y-axis variable.

Figure B.4: Event study illustration: controls (no compositional adjustment)



This figure repeats the exercise of Figure B.2 with controls depicted on the left y-axis, but without holding fixed the sample of states. The range of the left y-axis is 2 standard deviations of the left y-axis variable.

Figure B.5: Event study illustration: GSS attitude variables (no compositional adjustment)



This figure repeats the exercise of Figure B.3 with region-level GSS attitude variables depicted on the left y-axis, and without holding fixed the sample of states. The range of the left y-axis is 2 standard deviations of the left y-axis variable.

Table B.9: Do single moms migrate based on child support laws?

	(1)	(2)	(3)	(4)	(5)
Difference in CS. Laws	0.015 (0.021)	0.016 (0.022)	0.016 (0.022)	0.015 (0.022)	0.014 (0.023)
State f.e.		Yes	Yes	Yes	Yes
Year f.e.			Yes	Yes	Yes
State-year controls				Yes	Yes
Baseline controls * t					Yes
State-year f.e.					
Observations	13,594	13,594	13,594	13,402	12,707
R^2	0.017	0.024	0.026	0.027	0.025
Mean of dep. var.	0.03	0.03	0.03	0.03	0.03

The sample is from the March CPS and is restricted to never married mothers between 17 and 60 with a child 18 years or less in the household whose father is absent. This table estimates $Mig_{ist} = \beta_0 + \beta_1(CS_{s,t-1} - CS_{s',t-1}) + \gamma' x_{ist} + \alpha' w_{st} + \epsilon_{ist}$. The dependent variable Mig_{ist} is an indicator for whether the mother moved across U.S. states from year $t-1$ to t. The regressor of interest is the net benefit of migrating at time $t-1$, $(CS_{s,t-1} - CS_{s',t-1})$, where s is current state of residence, and s' is state of residence at time $t-1$. This regressor is standardized to have mean zero, standard deviation 1. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

C Additional Robustness Checks

C.1 Do social attitudes predict future child support statute adoption?

Here I explore whether social attitudes correlate with child support statute adoption. I regress the state-year child support index on region-year averages of social attitudes of 18-44 year olds sampled in the GSS. Each social attitude variable is standardized to have mean 0, standard deviation 1 in the distribution of region-year observations. Because of gaps in the GSS variables that occur in different years, there are only 384 observations in the regression sample; however, I wish to present the data as is, without imputation, to explore reduced-form correlations.

Table C.1 reports the results. In column (1) we see that region-years that are pro-abortion

legalization are more likely to adopt child support laws. Moving one standard deviation up in the region-year distribution of pro-abortion attitudes is correlated with a .033 increase in the child support index, a small but statistically significant effect; this effect goes away when state and year fixed effects are added. Similar correlations exist for other attitude variables: region-years more favorable to sex education in public schools are more likely to have child support laws; region-years that believe the government spends too much on welfare are less likely to have child support laws in place.[3] Adding region and year fixed effects reduces most of these effects to near zero and statistically insignificant. The exception is the belief that pre-marital sex is not wrong, which predicts less child support law adoption, but the effect size is quite small. The evidence taken together suggests social attitudes do not correlate strongly with law adoption once we remove region and year fixed effects.

[3]This last result may not be too surprising, since child support is often presented as a substitute for welfare, encouraging the absent parent to support single-parent families instead of the state. This relationship disappears, however, once region and year fixed effects are added to the regression, indicating that this correlation may reflect broad time trends, rather than sharp changes in attitudes that cause (or respond to) changes in child support laws.

Table C.1: Do social attitudes correlate with child support law adoption?

	Child support law index			
	(1)	(2)	(3)	(4)
Pro-abortion	0.033**	0.045**	0.013	0.015
legalization	(0.016)	(0.022)	(0.012)	(0.011)
Favor sex	0.067***	0.072***	0.021**	0.008
education in public schools	(0.019)	(0.023)	(0.008)	(0.007)
Divorce laws	-0.043**	-0.042*	-0.020*	-0.001
should be easier	(0.019)	(0.024)	(0.011)	(0.010)
Sex before	-0.028	-0.016	0.003	-0.022**
marriage not wrong	(0.022)	(0.020)	(0.016)	(0.009)
Spending too	-0.128***	-0.124***	0.019*	0.010
much on welfare	(0.014)	(0.015)	(0.010)	(0.007)
	0.004	0.036	0.013	-0.009
D-D-R	(0.055)	(0.059)	(0.031)	(0.021)
	0.152***	0.356***	0.028	0.009
D-R-D	(0.038)	(0.098)	(0.027)	(0.045)
	0.017	0.120	0.045	-0.011
D-R-R	(0.091)	(0.150)	(0.056)	(0.041)
	-0.029	0.110	-0.015	0.022
R-R-R	(0.060)	(0.101)	(0.034)	(0.044)
	0.065	0.030	0.081*	0.041
R-R-D	(0.073)	(0.124)	(0.046)	(0.048)
	0.007	0.213**	-0.073*	0.023
R-D-R	(0.055)	(0.101)	(0.042)	(0.046)
	0.118	0.231***	0.013	0.009
R-D-D	(0.080)	(0.062)	(0.079)	(0.045)
State f.e.		Yes		Yes
Year f.e.			Yes	Yes
Number of observations		384	384	384
R^2	0.314	0.479	0.817	0.928
Mean of dep. var.	0.50	0.50	0.50	0.50

The dependent variable is the state-year child support law index. Societal attitudes from the General Social Survey (GSS) are measured at the region-year and are standardized to have mean 0, standard deviation 1. State government party composition variables are ordered House-Senate-Governor (so that D-D-R, e.g., is a dummy variable indicating a Democratic House, Democratic Senate, and Republican Governor). Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

C.2 Placebo check: marriage analysis

Here I perform a placebo exercise that takes the baseline regression reported in Table 6, Column (4), and adds two regressors: an interaction of $nmpreg_{ist}$ with the 3-year *future* change in child support laws in the person's state-year, $\Delta CS_{st+} = CS_{s,t+4} - CS_{s,t+1}$, and the

variable ΔCS_{st+} itself. The motivation for this exercise is that future innovations in the law should not affect the probability of marriage – unless unobserved attitudes toward marriage are driving future legal changes, which would pose a threat to identification. The results are reported in Table C.2. Column (2) shows a positive and statistically insignificant effect. As an additional check, I add an interaction of $nmpreg_{ist}$ with 3 year *past* changes in the child support index, $\Delta CS_{st-} = CS_{s,t-1} - CS_{s,t-4}$, and include ΔCS_{st-} as a regressor itself. As expected, this column shows no significant effect on the probability of marriage conditional on pregnancy in time t. The placebo test mitigates a concern about reverse causality – there is no spurious correlation between marital behavior today and future changes in the law, nor with past changes in the law.

Table C.2: Placebo check: child support laws' effect on marriage (1977-1992)

	Baseline	Placebo (future C.S.)	Past C.S.
	(1)	(2)	(3)
N.M. Pregnant X C.S. Index	-0.118***	-0.100**	-0.121**
	(0.041)	(0.038)	(0.050)
N.M. Pregnant	0.381***	0.361***	0.393***
	(0.021)	(0.019)	(0.027)
C.S. Index	0.011**	0.003	0.005
	(0.005)	(0.007)	(0.005)
N.M. Pregnant X ΔCS$_{st+}$		0.101*	
		(0.052)	
ΔCS$_{st+}$		-0.007	
		(0.005)	
N.M. Pregnant X ΔCS$_{st-}$			-0.048
			(0.079)
ΔCS$_{st-}$			0.012*
			(0.006)
Soc. Attitudes Controls	Yes	Yes	Yes
State-Year Controls	Yes	Yes	Yes
Baseline controls * t	Yes	Yes	Yes
Number of individuals	53,000	48,000	50,500
R^2	0.071	0.080	0.067
Mean of dep. var.	0.06	0.05	0.06

Data is from 1992, 1993 and 1996 SIPP, and the sample is restricted to females in the 18-44 age cohort. The dependent variable is an indicator for the mother getting married for the first time in year t or $t+1$. Column (1) reports the baseline specification, Table 4 column (4). Column (2) adds regressors $nmpreg_{ist} \times \Delta CS_{s,t+}$ and $\Delta CS_{s,t+}$, while column (3) adds regressors $nmpreg_{ist} \times \Delta CS_{s,t-}$ and $\Delta CS_{s,t-}$. For a description of controls, see the footnote to Table 6. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table C.3: Placebo check (in levels): child support laws' effect on marriage (1977-1992)

	Baseline	Placebo (future C.S.)	Past C.S.
	(1)	(2)	(3)
N.M. Pregnant X C.S. Index	-0.118***	-0.224***	-0.153***
	(0.041)	(0.060)	(0.045)
N.M. Pregnant	0.381***	0.368***	0.394***
	(0.021)	(0.023)	(0.025)
C.S. Index	0.011**	0.006	0.015*
	(0.005)	(0.006)	(0.007)
N.M. Pregnant X CS_{st+3}		0.097	
		(0.060)	
CS_{st+3}		-0.003	
		(0.005)	
N.M. Pregnant X CS_{st-3}			0.026
			(0.059)
CS_{st-3}			-0.010
			(0.007)
Soc. Attitudes Controls	Yes	Yes	Yes
State-Year Controls	Yes	Yes	Yes
Baseline controls * t	Yes	Yes	Yes
Number of individuals	53,000	49,500	50,500
R^2	0.071	0.078	0.067
Mean of dep. var.	0.06	0.06	0.06

Data is from 1992, 1993 and 1996 SIPP, and the sample is restricted to females in the 18-44 age cohort. The dependent variable is an indicator for the mother getting married for the first time in year t or $t+1$. Column (1) reports the baseline specification, Table 4 column (4). Column (2) adds regressors $nmpreg_{ist} \times CS_{s,t+3}$ and $CS_{s,t+3}$, while column (3) adds regressors $nmpreg_{ist} \times CS_{s,t-3}$ and $CS_{s,t-3}$. For a description of controls, see the footnote to Table 6. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

C.3 Fertility

Consider now the effect of child support on the probability of having a nonmarital pregnancy. Consider the full panel of never-married women aged 18-44, and let $nmpreg_{ist}$ denote an indicator for never-married woman i becoming pregnant for the first time in time t. I estimate the following regression:

$$nmpreg_{ist} = \beta_0 + \beta_1 CS_{st} + \gamma' x_{ist} + \alpha' w_{st} + \epsilon_{ist} \tag{C.1}$$

In Equation C.1, the coefficient of interest is β_1, which represents the effect of child support enforcement on a pregnancy, for never-married women. Note that the coefficient β_1 may capture both a displacement effect (i.e. delaying children until the woman is older), or an effect on total nonmarital births. Equation C.1 is estimated with state and year fixed effects, and with controls for baseline state-year demographics interacted with year fixed effects, to control for state-specific trends that may be correlated with the adoption of child support statutes.

Table C.4 reports estimates using the child support index on the right hand side. The main result is that child support laws have a negative effect on the probability of never-married women becoming pregnant, but the effect is not statistically significant. Moving from a regime with 0 of the child support laws in place to one with all 9 decreases the probability of pregnancy by about 0.3 to 1.5 percent. This result, while not significant, is consistent with a finding in the literature that child support laws reduce nonmarital births (Plotnick et al., 2007; Case, 1998; Aizer and McLanahan, 2006).

The small fertility effects estimated here may reflect the model's ambiguous predictions for fertility: while men who have a strong dislike for children will father fewer of them, women are more willing to have premarital sex without commitment and abort fewer pregnancies, which leads to more children.

Table C.4: Child support laws' effect on fertility

	Female age cohort				Less than H.S.	H.S. Degree
	(1)	(2)	(3)	(4)	(5)	(6)
Child support	-0.003	-0.004	-0.011	-0.015	-0.058*	-0.007
law index	(0.007)	(0.010)	(0.011)	(0.011)	(0.034)	(0.011)
H.S. Degree	-0.021***	-0.021***	-0.021***	-0.021***		
	(0.004)	(0.004)	(0.004)	(0.004)		
African-American	0.011***	0.011***	0.011***	0.011***	-0.029***	0.018***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.007)	(0.002)
Other	-0.006*	-0.006*	-0.007*	-0.007*	-0.031**	-0.004
race/ethnicity	(0.004)	(0.004)	(0.004)	(0.004)	(0.014)	(0.004)
State f.e.	Yes	Yes	Yes	Yes	Yes	Yes
Year f.e.	Yes	Yes	Yes	Yes	Yes	Yes
Soc. Attitudes Controls		Yes	Yes	Yes	Yes	Yes
State-Year Controls			Yes	Yes	Yes	Yes
Baseline controls * t				Yes		
State-year f.e.						
Number of individuals	31,500	31,500	31,500	31,500	3,500	27,500
R^2	0.012	0.013	0.013	0.013	0.042	0.012
Mean of dep. var.	0.04	0.04	0.04	0.04	0.06	0.04

Data is from 1992, 1993 and 1996 SIPP. Sample is restricted to never-married women without children who are in the marital age cohort 18-44. The dependent variable is an indicator for the mother having a nonmarital pregnancy for the first time in year t. For a description of controls, see the footnote to Table 6. Column (5) restricts the sample to the less than high school education group; column (6) restricts the sample to high school and more. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

C.4 Placebo check: abortion analysis

A placebo check runs all four regressions of Table 4 but with the additional variable $\Delta CS_{st,+3} = CS_{st+4} - CS_{st+1}$ on the right hand side. This tests whether future changes in child support laws predict current abortion rates, conditional on the current child support regime. If future child support laws predicted current abortion rates, it would suggest the presence of omitted variables – that unobserved determinants of current abortion levels are also behind trends in child support law adoption. Table C.5 provides evidence against this concern.

Table C.5: Placebo check: the effect of child support laws on abortions (1978-1992)

	(1)	(2)	(3)	(4)	(5)	(6)
CS Index	-3.992**	-3.918**	-3.242**	-2.635*	-2.611	-2.781*
	(1.812)	(1.780)	(1.539)	(1.360)	(1.586)	(1.598)
ΔCS_{t+3}	-0.798	-0.761	-0.499	-0.085	-0.305	-0.233
	(1.079)	(1.073)	(0.973)	(1.042)	(0.951)	(1.111)
Ab. attitudes control		Yes	Yes	Yes	Yes	Yes
State-Year Controls			Yes	Yes	Yes	Yes
Baseline controls * t				Yes	Yes	Yes
Ab. providers control					Yes	
Region * t						Yes
Number of observations	528	528	528	528	528	528
R^2	0.977	0.977	0.978	0.979	0.981	0.980
Mean of dep. var.	27.55	27.55	27.55	27.55	27.55	27.55

The dependent variable is number of abortions per 1000 women age 15-44, regressed on the child support law index and state and year fixed effects. Standard errors are clustered at the state level. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

D Shotgun marriages: additional descriptive analysis

This section presents additional descriptive analysis of shotgun marriages. I next investigate whether shotgun marriages differ from non-shotgun marriages in terms of marital happiness and divorce hazard rates. SIPP includes the year of divorce if the respondent's first marriage was terminated in a divorce, and hence it is possible to study divorce hazards over 3-year, 5-year, and 10-year time horizons. I also use data from the National Survey of Families and Households (NSFH), Wave 1, which surveyed respondents in 1987-88, and, like SIPP, has retrospective marriage and fertility histories.

The NSFH has the advantage of including self-reported measures of marital happiness and stability, measured once at the time of the survey (1987-1988). The NSFH does not include state of residence identifiers, and hence it is infeasible to link this data with state-year legislative data. I follow the work of Bertrand et al. (2015) and define three marital

quality variables based on three survey questions in the NSFH data. The first question asks "Taking things all together, how would you describe your marriage?" Respondents can choose answers from a scale of 1 (very unhappy) to 7 (very happy). I define a binary variable "very happy" as answering 7 to this question.[4] The second question asks if, during the past year, either the husband or wife has discussed the idea of separating. I define "discuss separation" as a binary variable taking the value of 1 if the respondent answers yes that either partner has discussed separation. The third question asks about trouble in the marriage: "During the past year, have you ever thought that your marriage might be in trouble?" and I define a variable "marriage trouble" equal to 1 if the respondent answers yes.

The NSFH selects one adult per household as the primary respondent, which is chosen at random at the time of sampling. I restrict the sample to primary respondents only, and restrict further to those who are either married, separated, or divorced, and aged 18 or over at the time of the survey. I estimate the following regression

$$q_{it} = \beta_0 + \beta_1 shotgun_{it} + \gamma' x_{it} + \epsilon_{it} \tag{D.1}$$

The dependent variable q_{it} represents marital quality using one of the measures discussed above. The $shotgun_{it}$ variable indicates if the marriage was a shotgun marriage. The vector of controls x_{it} includes dummies for the year of marriage, gender, race, completed years of schooling, number of children, and census region (geographic identifiers finer than census region are suppressed by the NSFH for confidentiality purposes). The vector x_{it} also includes a variable for household income and respondent's age at the time of the interview. Note that because the $shotgun_{it}$ variable is an imperfect empirical approximation to the theoretical definition given within the model, there is potential for measurement error to attenuate the estimate of β_1.

Since I only observe marital stability variables for still-existing marriages, there is a

[4]About 46 percent of survey respondents answer "very happy" and hence this represents a natural division of responses into a binary variable; the results are not sensitive to this choice of definition.

selection concern because the lowest quality marriages will not be in the sample. Hence I impute values for marriages that ended in separation or divorce: I code "very happy" equal to 0, "discuss separation" equal to 1, and "marriage trouble" for these marriages.[5] Using the imputed data, 34 percent of respondents report their marriage as "very happy"; 46 percent of the sample answers yes to discussing separation with their spouse; and 28 percent of the sample reports that in the past year they thought their marriage might be in trouble.

Table D.1 reports estimates of estimation of Equation D.1 using NSFH data. First marriages that are shotgun marriages are 7.2 percent less likely to be reported by the respondents as "very happy". In addition, shotgun marriage respondents are 6.2 percent more likely to discuss the idea of separating with their partner, and 4.2 percent more likely to report having thought the marriage might be in trouble in the last year. I estimate Equation D.1 using only the selected sample of marriages that remain intact as of the time of sampling (i.e. without any imputation for divorced couples), and report the results in Table D.2. The results are similar: shotgun-married couples are less likely to report being very happy, more likely to discuss separation, and more likely to report their marriage being in trouble.

Table D.3 reports the results from estimating Equation D.1 using SIPP with a dummy for divorce over varying lengths of time as the dependent variable. The 3-year divorce hazard is defined as a binary variable taking a value of 1 if the respondent was divorced within 3 years of her first marriage, 0 otherwise. The 5-year and 10-year hazards are defined similarly for the 5- and 10-year intervals, respectively. For this regression, I restrict the sample to marriages occurring between 1975 and h years prior to the SIPP survey year, where h is the hazard interval being studied in the regression. This is to ensure that marriages in the regression sample do not have 0 for the h-year divorce hazard by construction. Table D.3 shows that there is no detectable increase in the probability of divorce within the first three years of marriage for a shotgun marriage relative to a non-shotgun marriage. A shotgun marriage has a 2.0 to 2.1 percent higher probability of divorce within the first 5 years (off of

[5]I report estimates without imputation as well, i.e. using the selected sample of marriages that remain as of the NSFH survey date. (See Table D.2.)

a baseline of 10 percent) although it is not significant, and a 3.4 percent higher probability of divorce over a 10-year period (off of a baseline of 28 percent).

Table D.1: Match quality of shotgun marriages relative to other marriages

	Very happy	Discuss Separation	Marriage trouble
	(1)	(2)	(3)
Shotgun marriage	-0.072***	0.062***	0.042***
	(0.017)	(0.022)	(0.015)
Female	-0.041***	-0.017	0.042***
	(0.011)	(0.015)	(0.010)
African-American	-0.115***	0.131***	0.110***
	(0.019)	(0.022)	(0.018)
Other	-0.006	0.057**	-0.042**
race/ethnicity	(0.021)	(0.027)	(0.019)
No children	0.007	0.028	0.002
	(0.018)	(0.023)	(0.016)
Number of	0.007**	-0.017***	-0.011***
children	(0.004)	(0.005)	(0.003)
Age at survey	-0.006	-0.092***	-0.023
	(0.017)	(0.023)	(0.016)
North Central	-0.010	0.025	0.002
	(0.016)	(0.021)	(0.014)
South	0.048***	0.025	-0.008
	(0.015)	(0.020)	(0.013)
West	0.001	0.041*	0.044***
	(0.017)	(0.022)	(0.015)
Log Household	0.050***	-0.064***	-0.064***
Inc.	(0.005)	(0.007)	(0.005)
HS Degree	-0.026*	0.010	0.004
	(0.015)	(0.019)	(0.014)
Some college	-0.085***	0.053**	0.041**
	(0.018)	(0.023)	(0.016)
College degree	-0.088***	0.093***	0.048***
	(0.019)	(0.025)	(0.017)
Marriage yr. dummies	Yes	Yes	Yes
Observations	8,343	4,508	8,343
R^2	0.045	0.195	0.099
Mean of dep. var.	0.34	0.46	0.28

Results from the estimation of Equation D.1. Data is from 1987-1988 NSFH, and the sample consists of married respondents age 18 and over. Each column has as its dependent variable one of the three marital happiness variables defined in the text. Standard errors in parentheses. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table D.2: Match quality of shotgun marriages relative to other marriages conditional on staying married

	Very happy	Discuss Separation	Marriage trouble
	(1)	(2)	(3)
Shotgun marriage	-0.099***	0.090**	0.039**
	(0.025)	(0.042)	(0.019)
Female	0.029*	-0.019	0.017
	(0.016)	(0.028)	(0.012)
African-American	-0.060*	0.071	-0.005
	(0.031)	(0.049)	(0.024)
Other race/ethnicity	-0.022	0.036	-0.052**
	(0.029)	(0.052)	(0.023)
No children	0.055**	-0.015	-0.065***
	(0.027)	(0.048)	(0.021)
Number of children	0.004	-0.011	-0.001
	(0.006)	(0.011)	(0.004)
Age at survey	-0.027	-0.031	0.011
	(0.026)	(0.046)	(0.020)
North Central	-0.031	0.026	0.014
	(0.022)	(0.041)	(0.017)
South	0.032	0.068*	0.029*
	(0.021)	(0.039)	(0.016)
West	-0.008	0.088**	0.045**
	(0.024)	(0.043)	(0.019)
Log Household Inc.	-0.002	-0.021	-0.003
	(0.009)	(0.015)	(0.007)
HS Degree	-0.039*	0.025	0.010
	(0.023)	(0.042)	(0.018)
Some college	-0.090***	0.023	0.022
	(0.027)	(0.048)	(0.021)
College degree	-0.093***	0.110**	0.027
	(0.027)	(0.051)	(0.021)
Marriage yr. dummies	Yes	Yes	Yes
Observations	4,288	1,281	4,288
R^2	0.057	0.124	0.071
Mean of dep. var.	0.43	0.31	0.17

Results from the estimation of Equation D.1. Data is from 1987-1988 NSFH. Each column has as its dependent variable one of the three marital happiness variables defined in the text. The sample consists of respondents 18 and over whose first marriage is intact at the survey date. Standard errors in parentheses. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$

Table D.3: Shotgun marriages and the likelihood of divorce

	Divorce (3yr)		Divorce (5yr)		Divorce (10yr)	
	(1)	(2)	(3)	(4)	(5)	(6)
Shotgun marriage	0.003	0.004	0.019*	0.020*	0.062***	0.061***
	(0.006)	(0.007)	(0.010)	(0.010)	(0.018)	(0.017)
African-American	-0.000	-0.003	0.010	0.010	0.052**	0.048**
	(0.006)	(0.007)	(0.007)	(0.007)	(0.020)	(0.020)
Other	-0.030***	-0.026**	-0.030**	-0.026**	-0.079***	-0.077***
race/ethnicity	(0.009)	(0.010)	(0.011)	(0.012)	(0.016)	(0.016)
Age at first	-0.005***	-0.005***	-0.011***	-0.010***	-0.018***	-0.018***
mar.	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
HS Degree	-0.010	-0.010	-0.018	-0.017	-0.040	-0.035
	(0.007)	(0.009)	(0.016)	(0.016)	(0.025)	(0.024)
Some college	-0.004	-0.004	0.005	0.005	0.028	0.031
	(0.007)	(0.011)	(0.017)	(0.017)	(0.023)	(0.023)
College degree	-0.041***	-0.039***	-0.056***	-0.054***	-0.067**	-0.062**
	(0.007)	(0.011)	(0.019)	(0.018)	(0.027)	(0.027)
Marriage yr. dummies	Yes	Yes	Yes	Yes	Yes	Yes
State f.e.	No	Yes	No	Yes	No	Yes
Observations	20,500	20,500	18,500	18,500	7,000	7,000
R^2	0.024	0.030	0.048	0.055	0.087	0.097
Mean of dep. var	0.09	0.09	0.16	0.16	0.28	0.28

Results from the estimation of Equation D.1. Data is from 1992, 1993 and 1996 restricted-SIPP. The dependent variables are divorce hazard in the first 3 years, first 5 years, and any divorce. Standard errors in parentheses. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$